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(54) QUAD FLAT NO-LEAD (QFN) PACKAGE WITH TIE BARS AND DIRECT CONTACT INTERCONNECT BUILD-UP STRUCTURE AND METHOD FOR MAKING THE SAME

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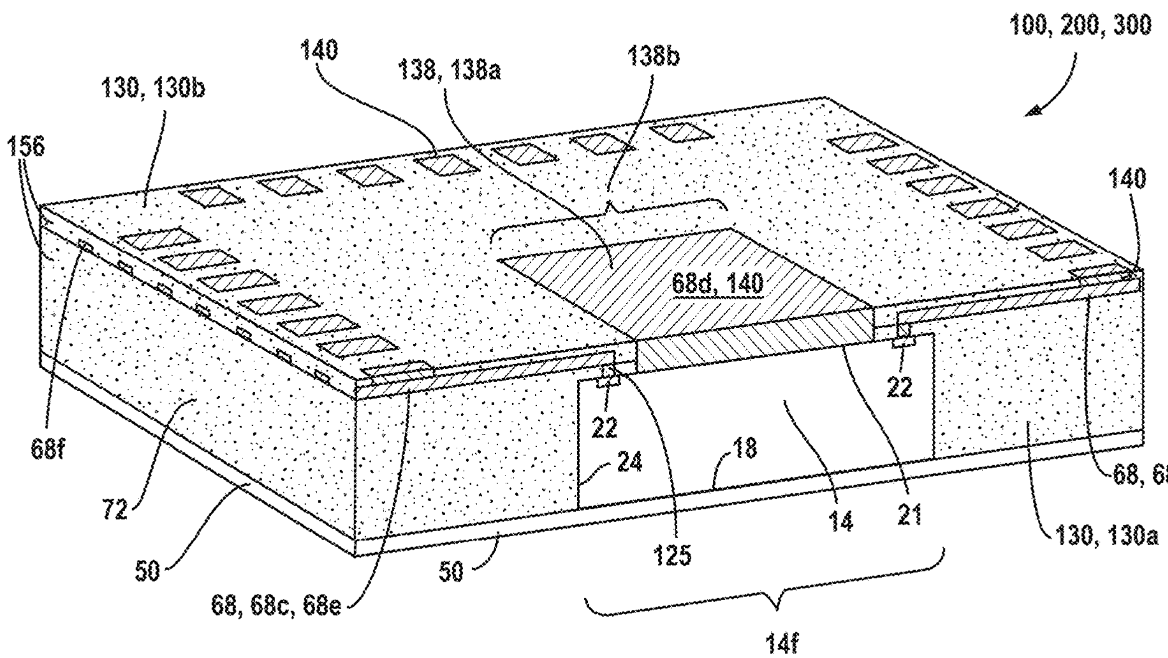
(57) **ABSTRACT**

A method of making a QFN, DEN, or SON package may comprise forming an embedded chip panel comprising a plurality of semiconductor chips embedded in encapsulant and separated by saw streets. A conductive layer may be formed over the embedded chip panel, the conductive layer comprising bussing lines, contact pads, traces, and tie bars, wherein the bussing lines and tie bars extend into the saw streets. Vertical conductive elements may be disposed over and coupled with the conductive layer. An encapsulant layer may be disposed over the conductive layer and around the vertical conductive elements, wherein the vertical conductive elements comprise exposed ends and at least a portion of the tie bars is exposed from the encapsulant layer. Land pads, a conductive pad finish, or SMS may be electroplated over the vertical conductive elements by providing a current through the bussing lines and tie bars.



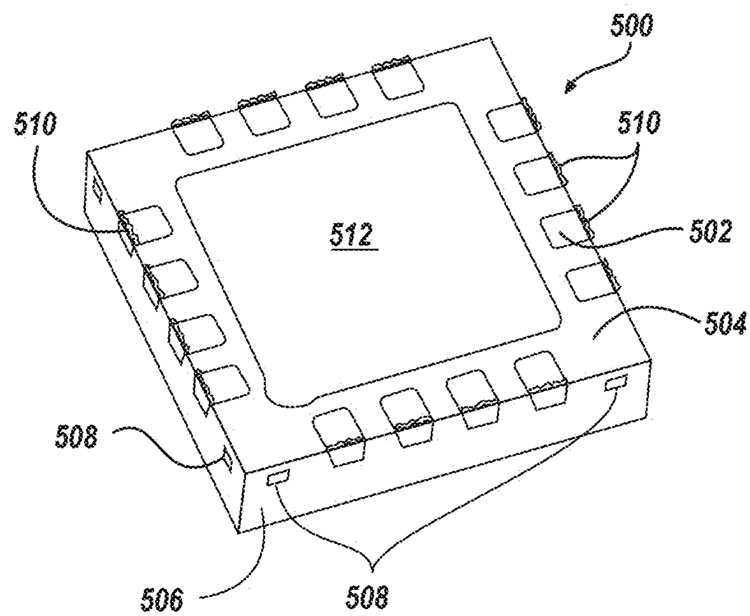


FIG. 1A - Prior Art

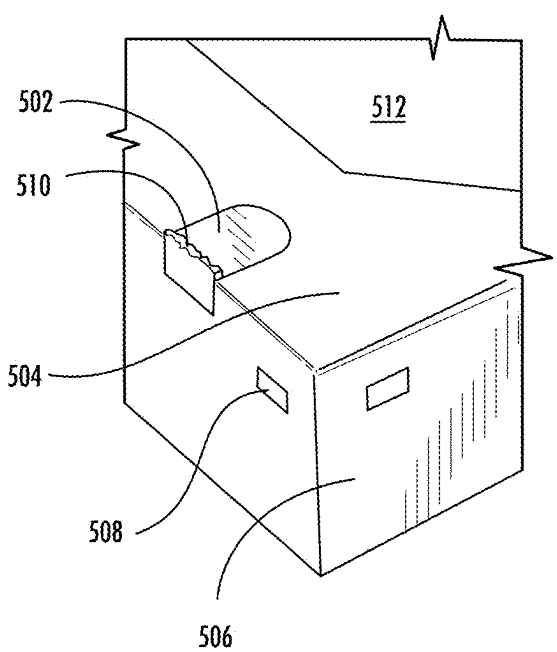


FIG. 1B - Prior Art

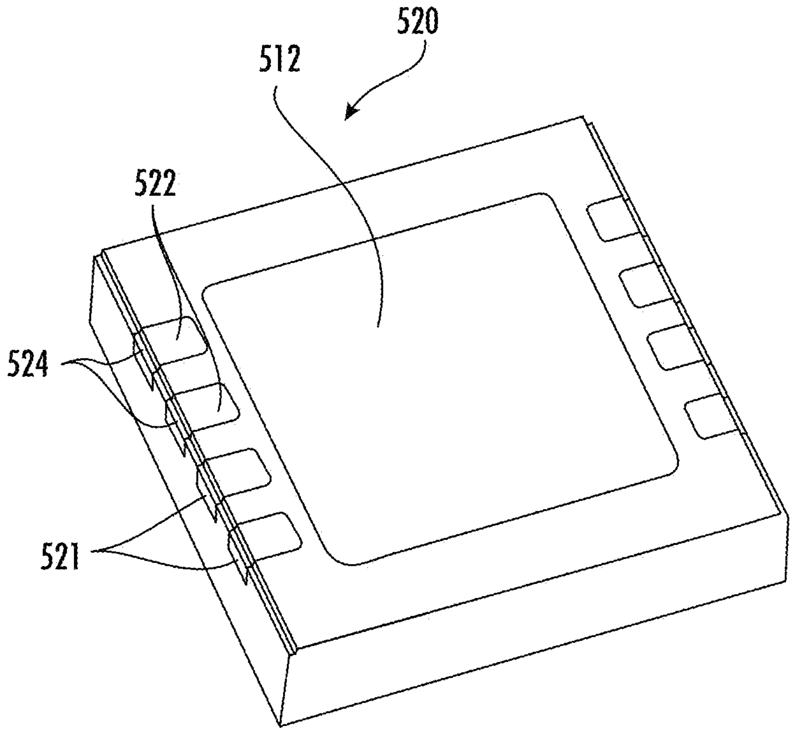


FIG. 1C - Prior Art

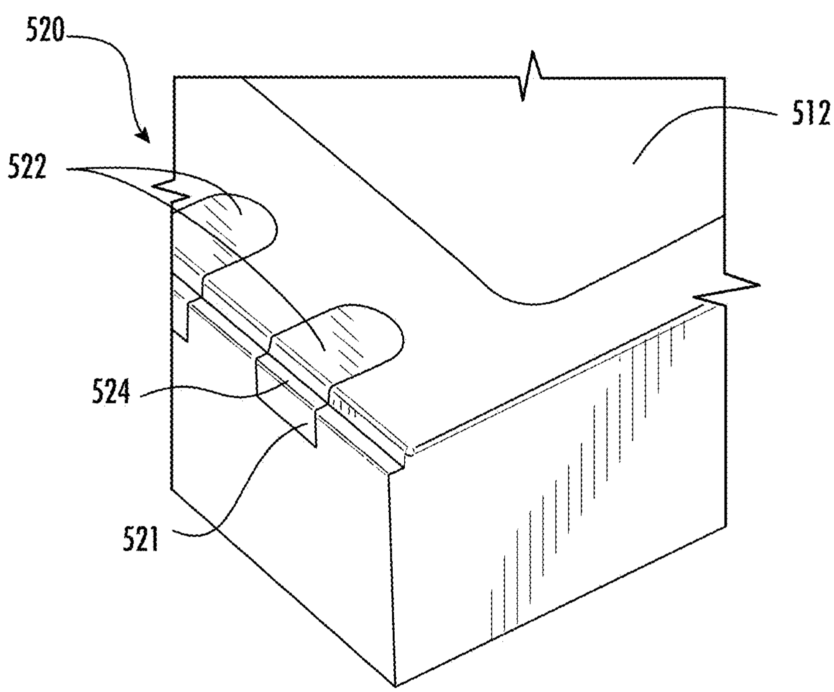


FIG. 1D - Prior Art

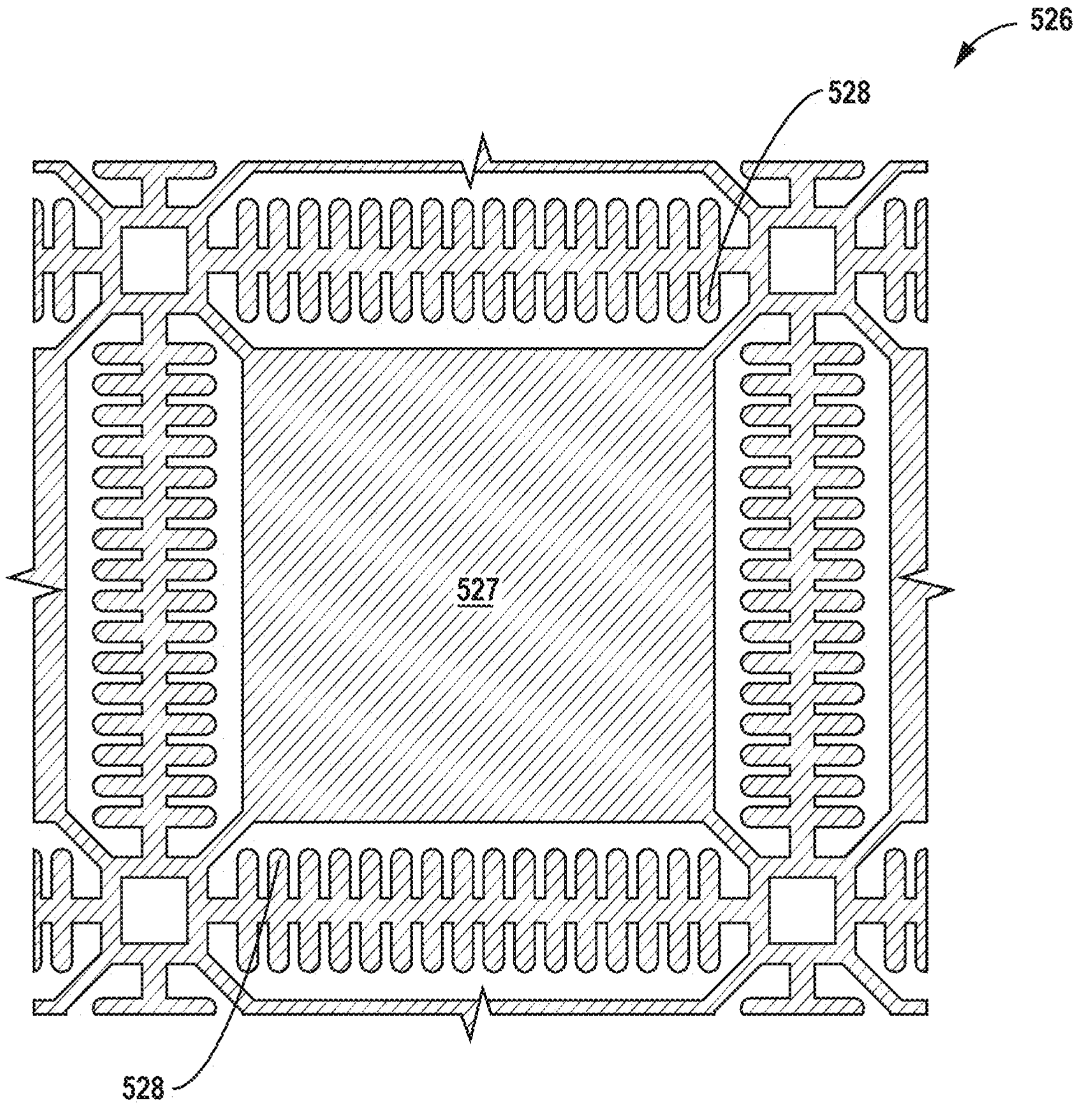
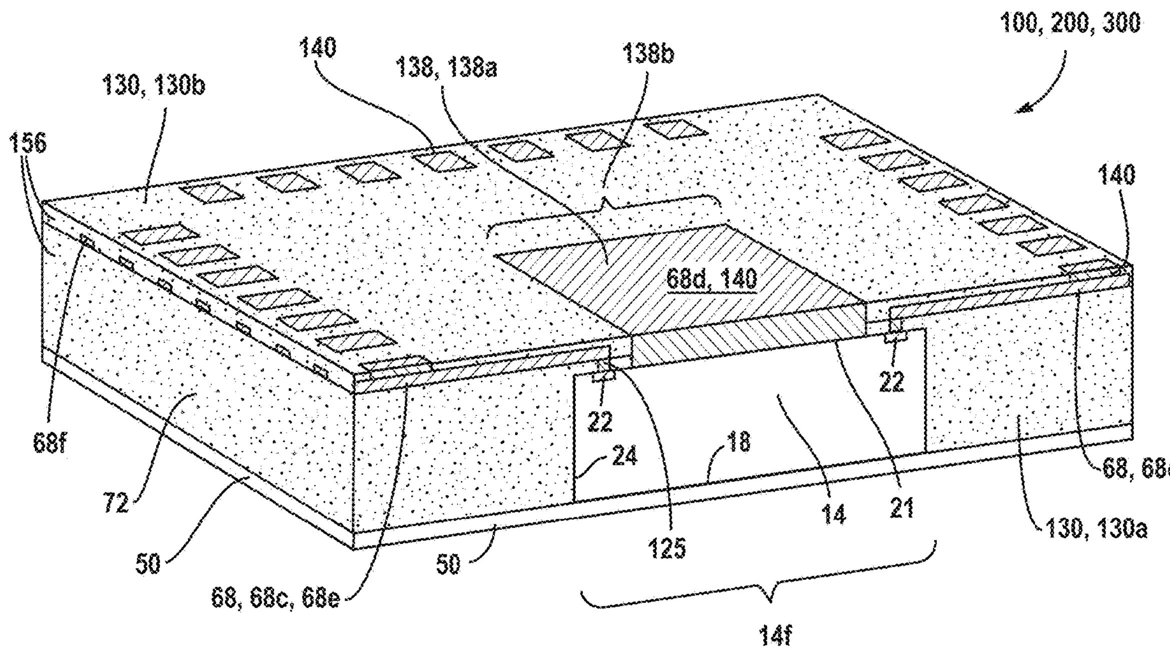


FIG. 1E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138, 138a
138b
130, 130b
156
140
68d, 140
22
22
68f
24
18
72
14
21
68, 68c
125
130, 130a
50
50
68, 68c, 68e
14f

FIG. 2

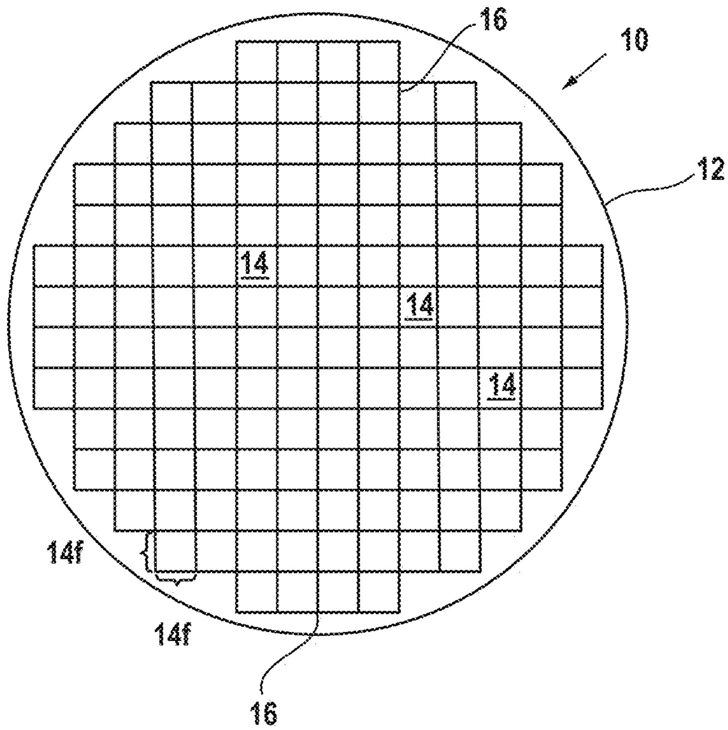


FIG. 3A

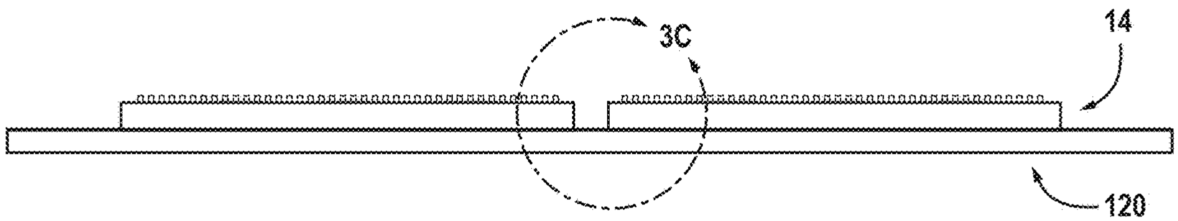


FIG. 3B

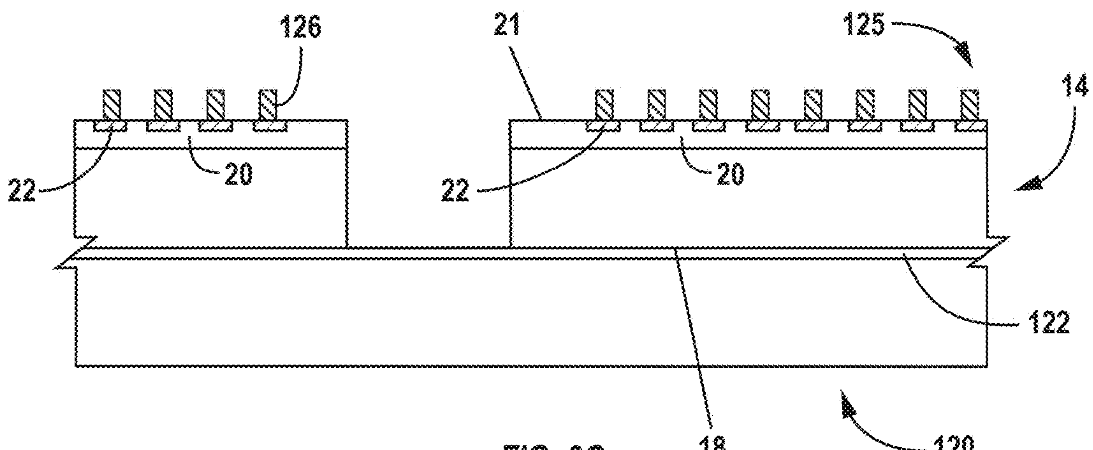


FIG. 3C

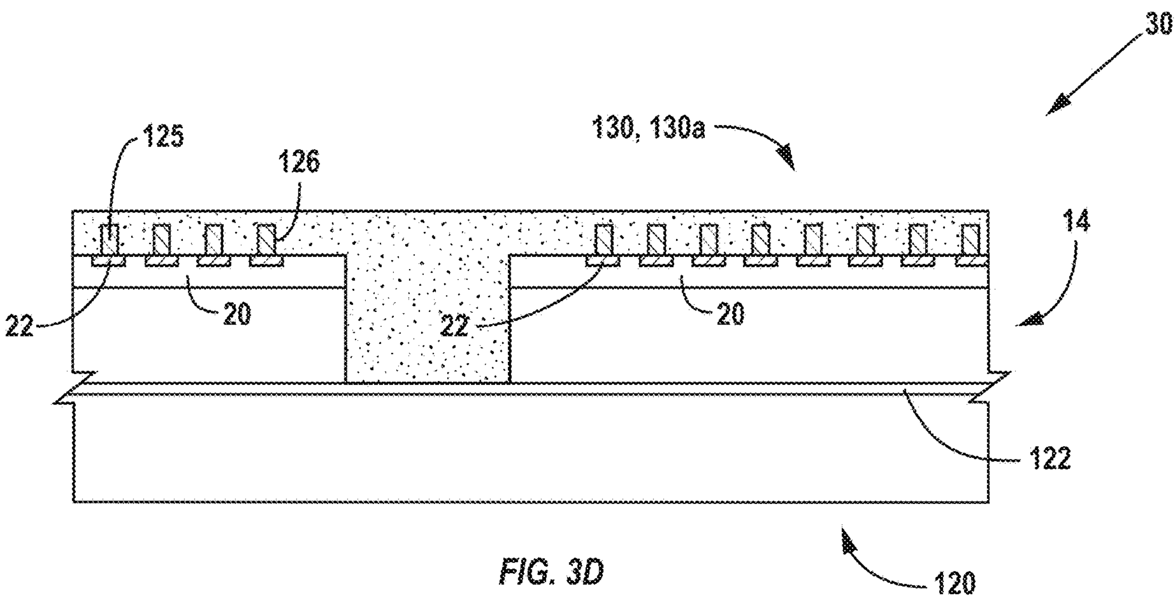


FIG. 3D

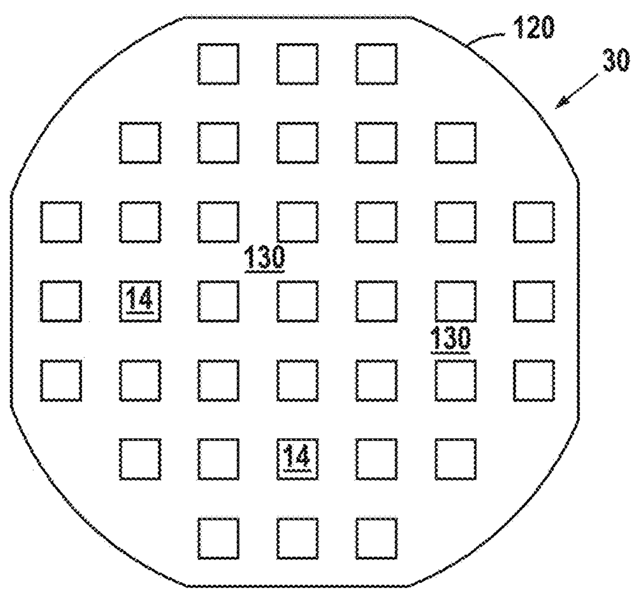


FIG. 3E

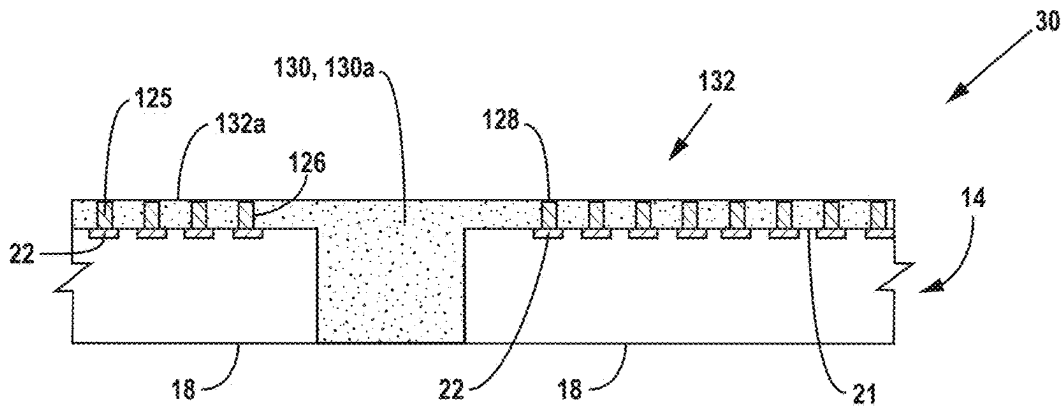
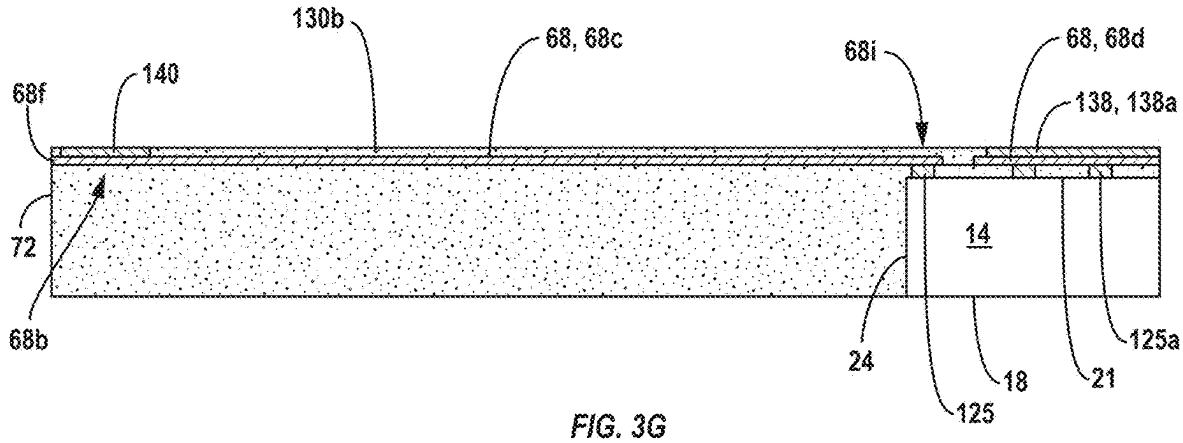

30
130, 130a
132
125
132a
128
126
14
22
18
22
18
21

FIG. 3F


130b
68, 68c
68i
68, 68d
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138, 138a
68f
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24
18
21
125a
125

FIG. 3G

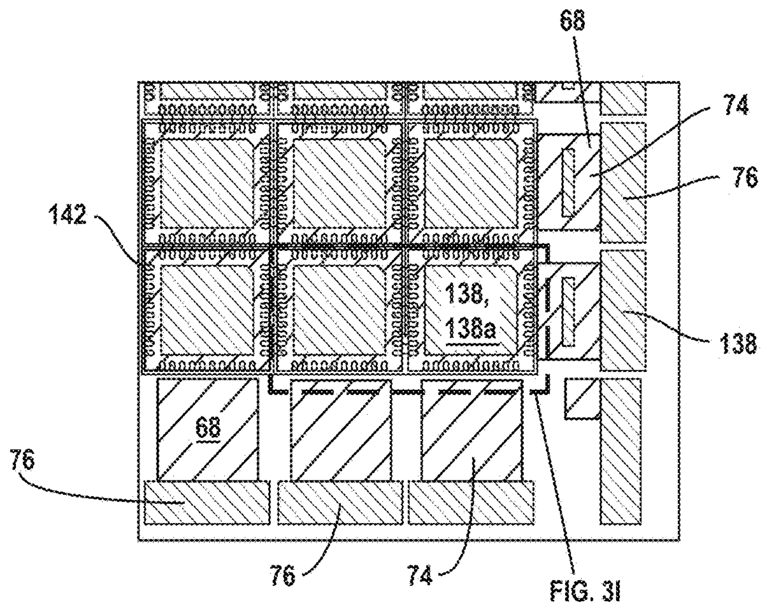

68
74
76
142
138,
138a
138
68
76
76
74
FIG. 3I

FIG. 3H

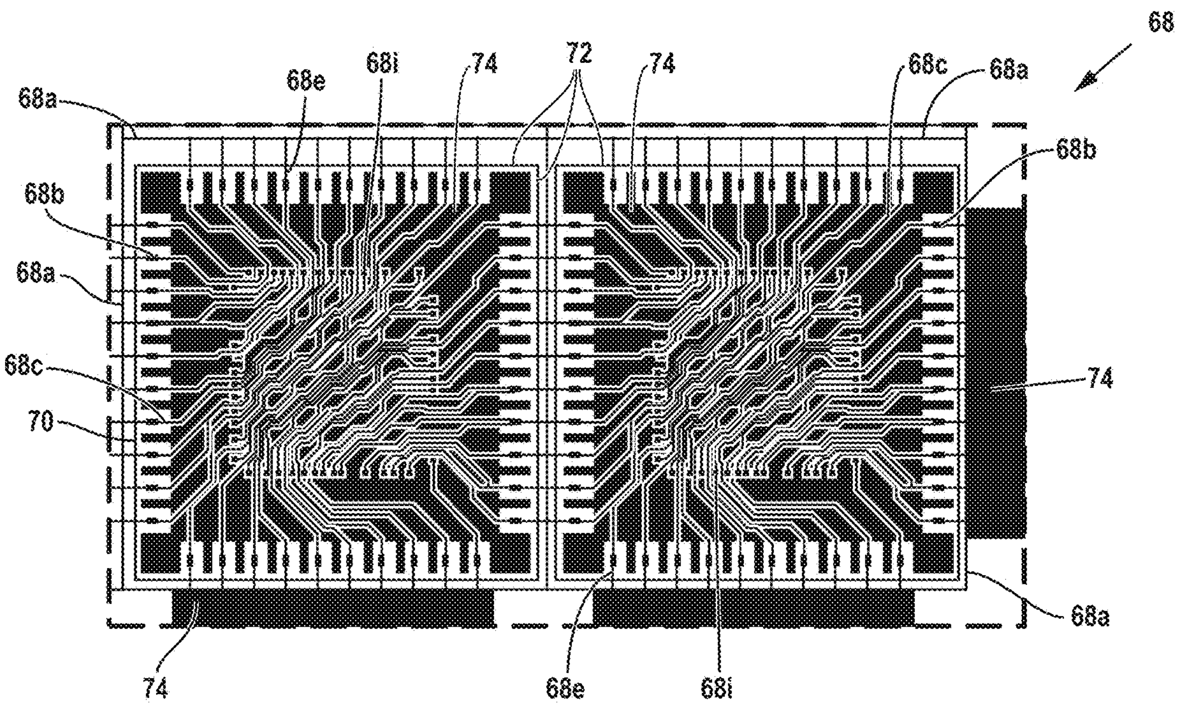
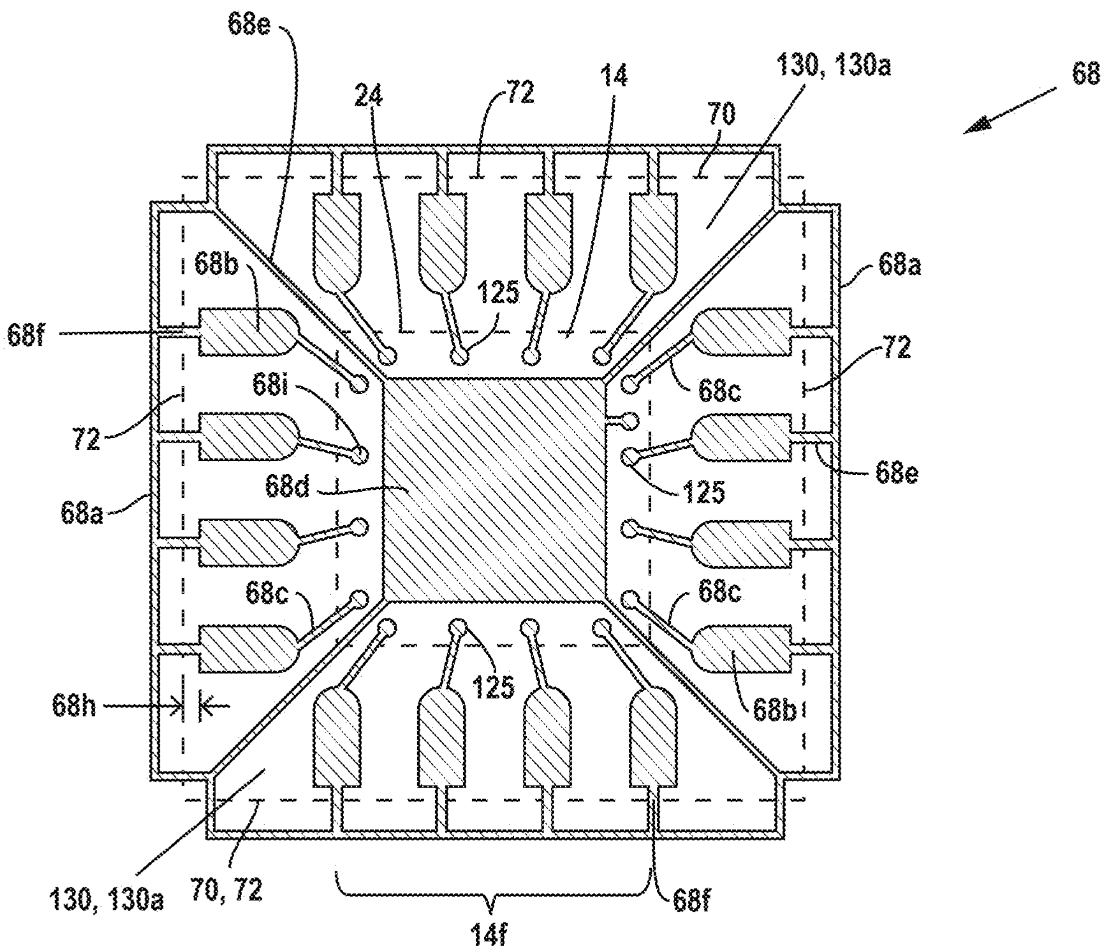

68
68a
68e
68i
74
72
74
68c
68a
68b
68b
68a
68c
70
74
68a
74
68e
68i

FIG. 3I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68b
125
68a
68f
72
68c
68i
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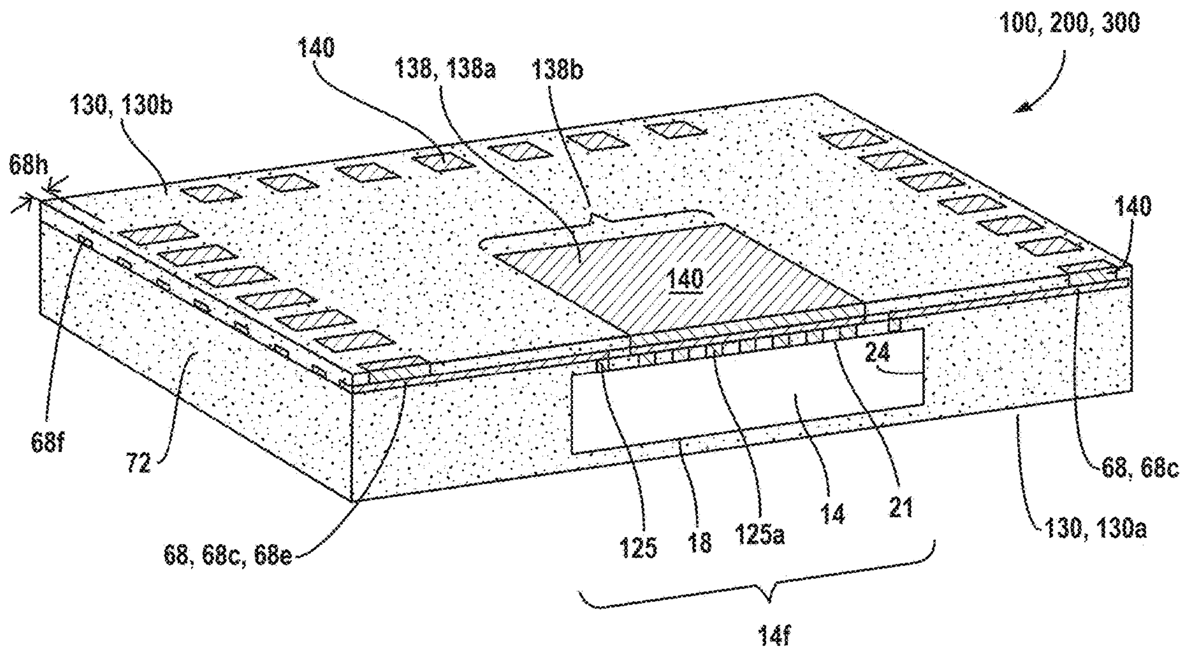
68e
68d
68a
68c
68c
125
68b
68h
130, 130a
70, 72
68f
14f

FIG. 3J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68, 68c, 68e
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18
125a
14
21
14f

FIG. 3K

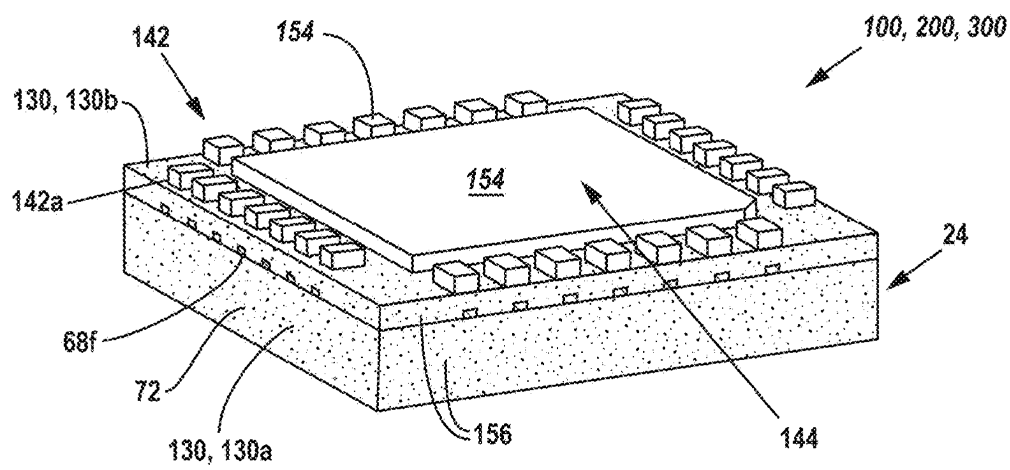


FIG. 4A

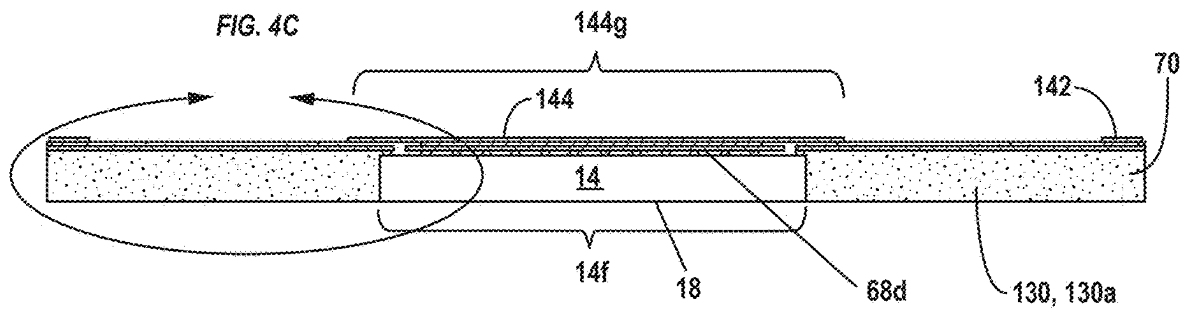


FIG. 4B

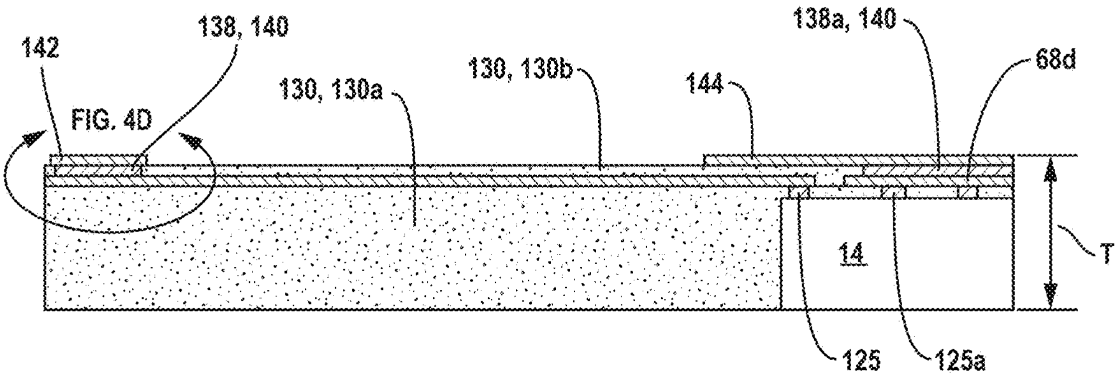


FIG. 4C

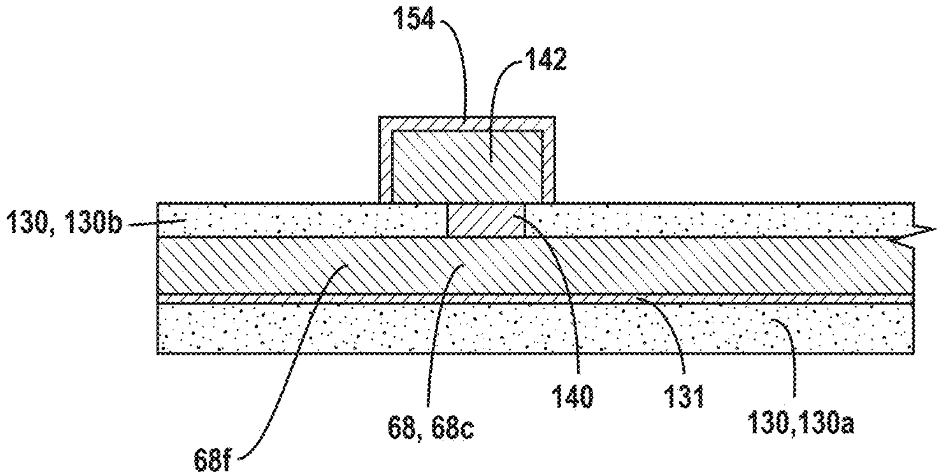


FIG. 4D$_1$

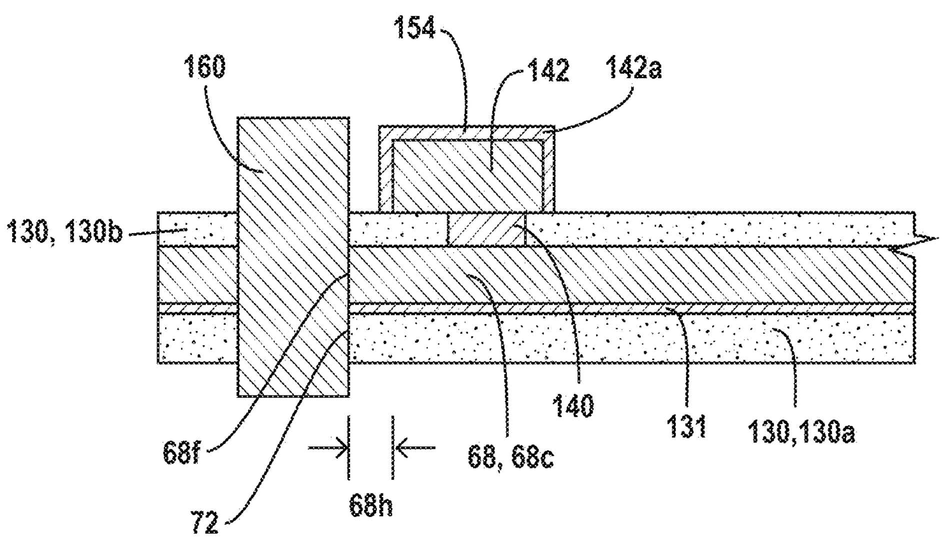


FIG. 4D$_2$

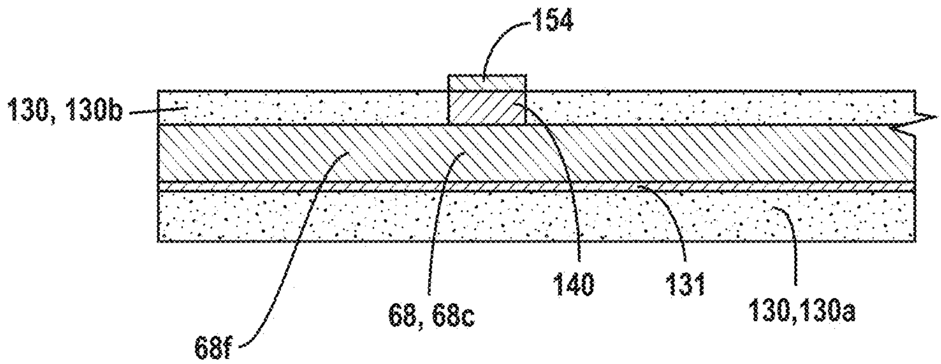


FIG. $4E_1$

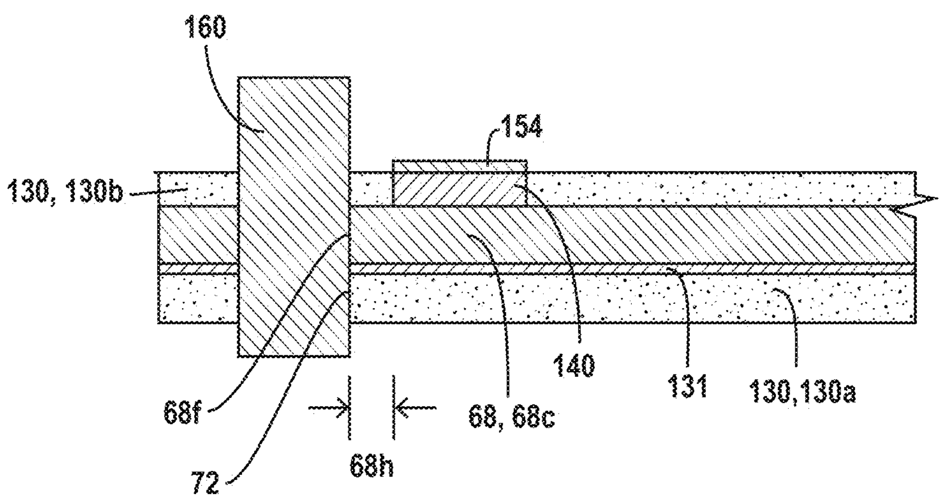


FIG. $4E_2$

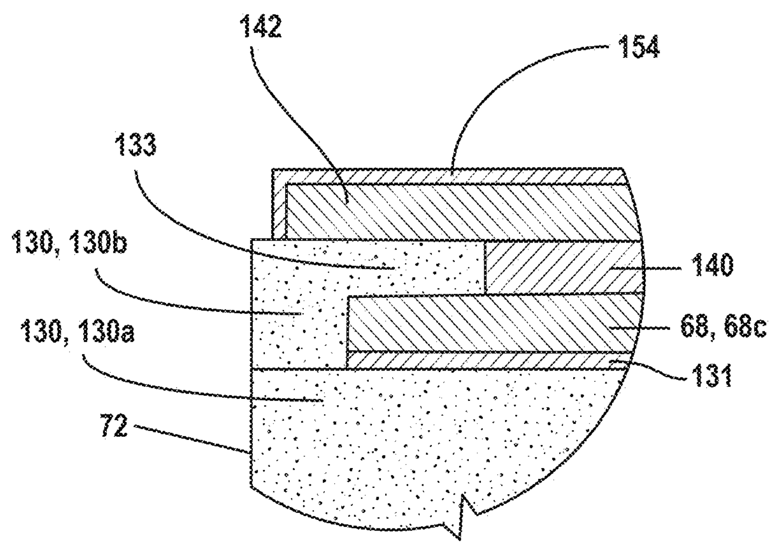


FIG. 4F

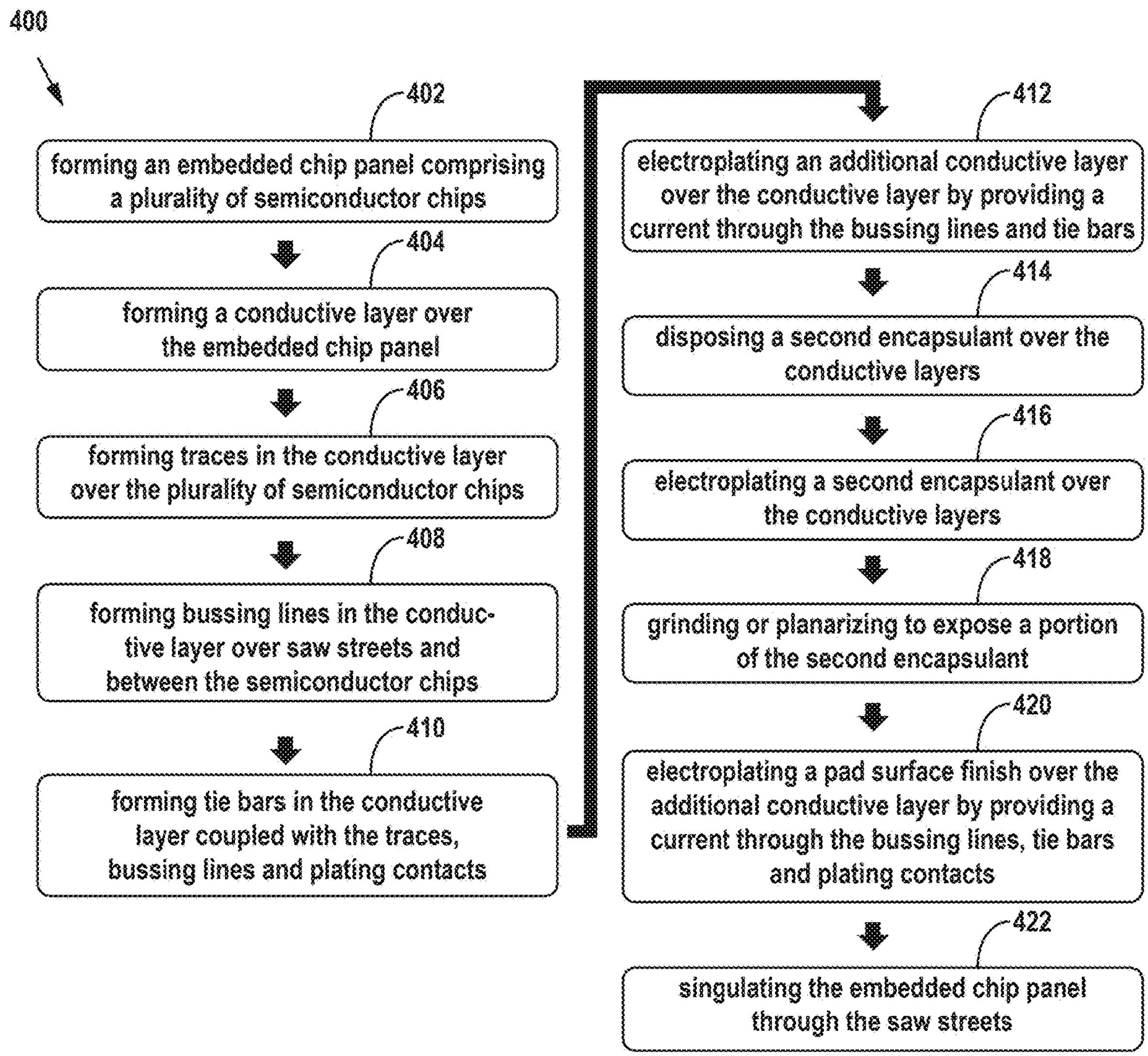
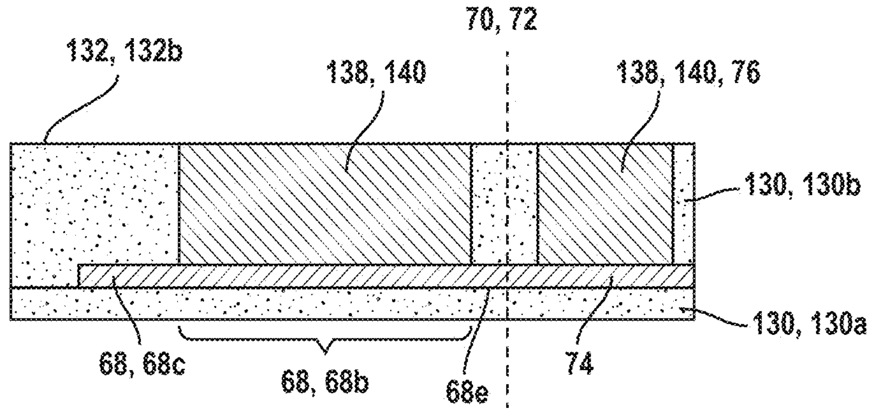

400
402
forming an embedded chip panel comprising a plurality of semiconductor chips
404
forming a conductive layer over the embedded chip panel
406
forming traces in the conductive layer over the plurality of semiconductor chips
408
forming bussing lines in the conductive layer over saw streets and between the semiconductor chips
410
forming tie bars in the conductive layer coupled with the traces, bussing lines and plating contacts
412
electroplating an additional conductive layer over the conductive layer by providing a current through the bussing lines and tie bars
414
disposing a second encapsulant over the conductive layers
416
electroplating a second encapsulant over the conductive layers
418
grinding or planarizing to expose a portion of the second encapsulant
420
electroplating a pad surface finish over the additional conductive layer by providing a current through the bussing lines, tie bars and plating contacts
422
singulating the embedded chip panel through the saw streets

FIG. 5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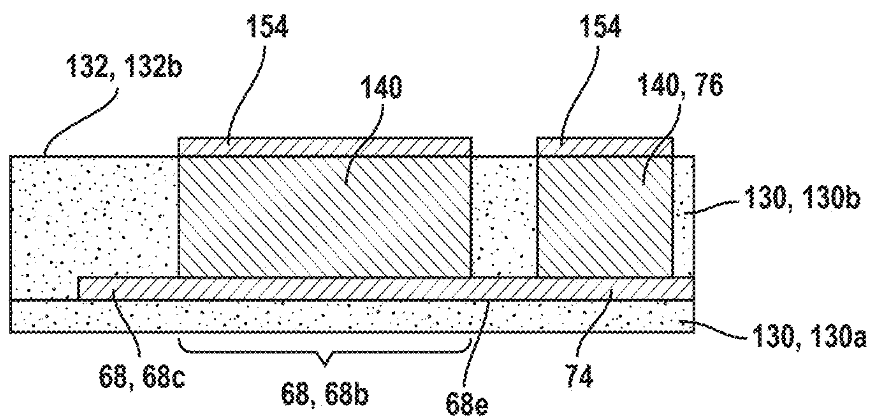
68e
74

FIG. 6A


154
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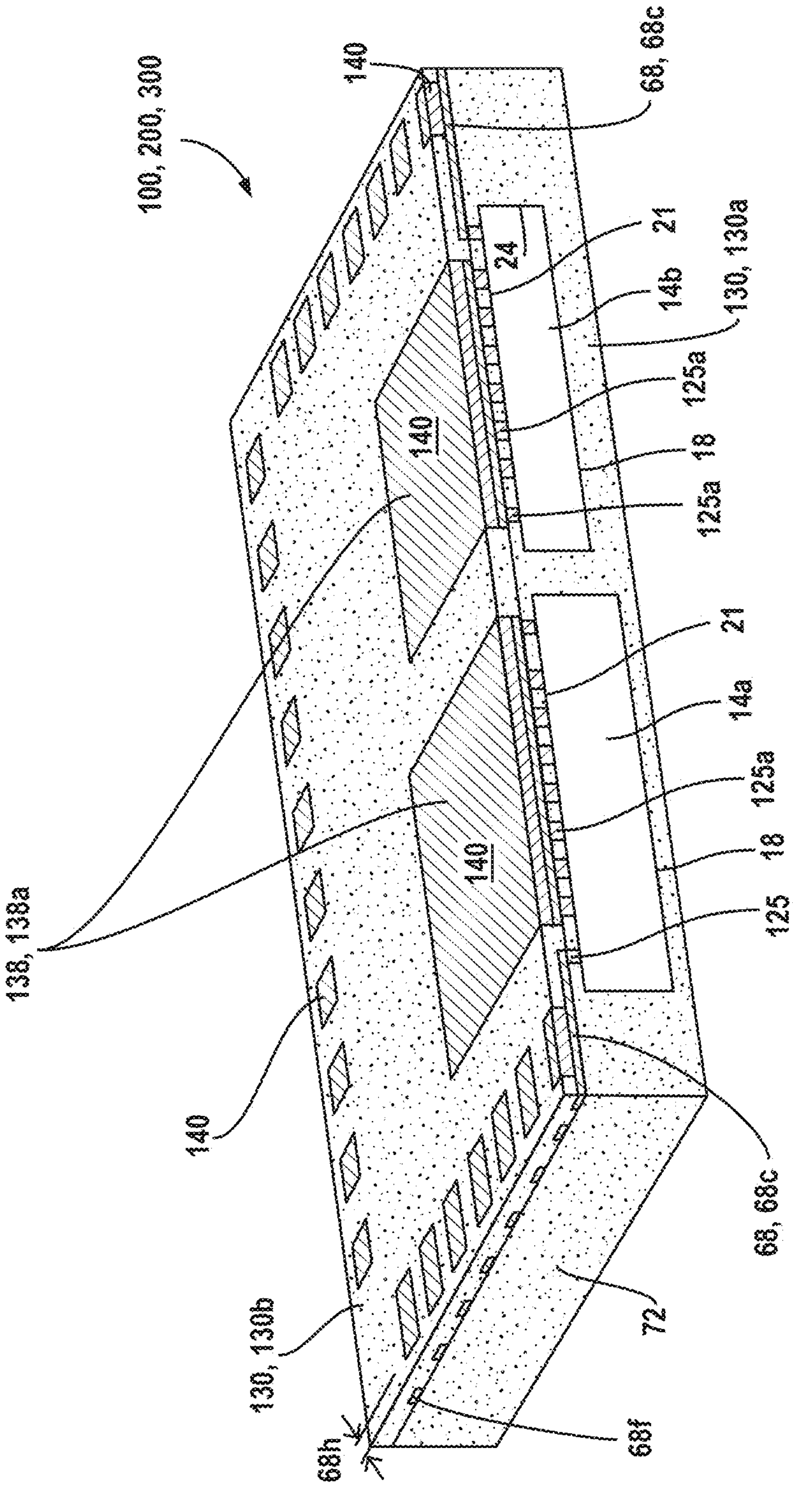
130, 130b
130, 130a
68, 68c
68, 68b
68e
74

FIG. 6B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138, 138a
140
68, 68c
72
130, 130b
68f
68h

FIG. 7

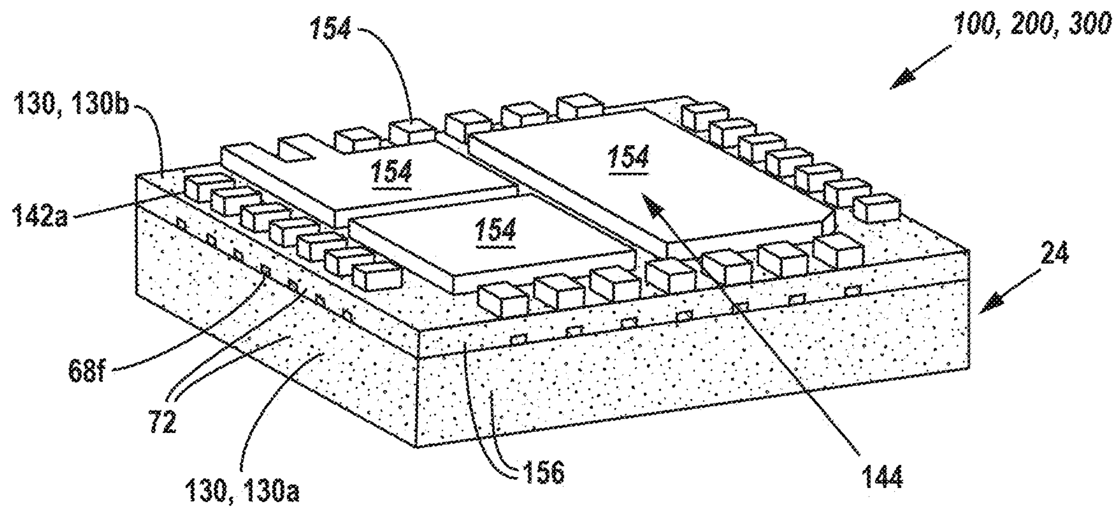
154
100, 200, 300
130, 130b
154
154
154
142a
24
68f
72
130, 130a
156
144

FIG. 8

QUAD FLAT NO-LEAD (QFN) PACKAGE WITH TIE BARS AND DIRECT CONTACT INTERCONNECT BUILD-UP STRUCTURE AND METHOD FOR MAKING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Patent No. 63/560,525, entitled "Quad Flat No-Lead (QFN) Package with Tie Bars and Direct Contact Interconnect Build-Up Structure and Method for Making the Same," filed on Mar. 1, 2024, the entire disclosure of which is hereby incorporated herein by this reference.

INCORPORATION BY REFERENCE

[0002] This disclosure incorporates by reference the entirety of the disclosures of: (i) U.S. patent application Ser. No. 13/891,056 filed May 9, 2013 and entitled "Semiconductor Device and Method of land Grid Array Packaging with Bussing Lines, now U.S. Pat. No. 9,269,622, and (ii) U.S. patent application Ser. No. 17/957,683 filed Sep. 30, 2022 and entitled "Quat Flat No-lead (QFN) Package without Leadframe and Direct Contact Interconnect Build-up Structure and Method for Making the Same."

TECHNICAL FIELD

[0003] This disclosure concerns devices and methods of forming quad flat no-lead (QFN), dual flat no-lead (DFN) or small-outline no-lead (SON) semiconductor packaging with tie bars and with molded direct contact interconnect build-up structures.

BACKGROUND

[0004] Semiconductor devices, packages, substrates, and interposers are commonly found in modern electronic products. Production of semiconductor devices involves a multistep build-up of components. Conventional interconnect structures alternate dielectric and conductive layers. An opening or via is created in the dielectric to allow connectivity from one layer to another. On the conductive layers, capture pads are required for the vias to correct for inconsistencies in manufacture. Use of these conventional capture pads impacts the ability to construct compact structures due to limits on routing density. Additionally, traditional manufacturing processes involve the use of leadframes that result in exposed leadframe ends on the sides of the packaging.

SUMMARY

[0005] In some aspects, the disclosure relates to a method of making a QFN, DFN, SON package comprising forming an embedded chip panel comprising a plurality of semiconductor chips embedded in encapsulant and separated by saw streets. A conductive layer may be formed over the embedded chip panel, the conductive layer comprising: contact pads and traces formed over the plurality of semiconductor chips, the traces extending beyond an edge of the semiconductor chips and coupled with vertical conductive elements; bussing lines formed over the saw streets; and tie bars extending between and coupled with the contact pads and the bussing lines. Vertical conductive elements may be formed over and coupled with the conductive layer. An encapsulant layer may be disposed over the conductive layer and around the vertical conductive elements such that the vertical conductive elements comprise exposed ends, and wherein at least a portion of the tie bars comprise cut tie bar ends exposed from the encapsulant layer. Land pads, a conductive pad finish, or a SMS may be electroplated over the vertical conductive element by providing a current through the bussing lines and tie bars, wherein the land pads, conductive pad finish, or SMS are electroplated and protrude from the encapsulant layer. The embedded chip panel may be singulated through the saw streets to form a plurality of QFN, DFN, or SON packages with exposed ends of the tie bars exposed at a periphery of the QFN, DEN, or SON packages.

[0006] In an aspect of the disclosure the method may further comprise forming the embedded chip panel by disposing a plurality of semiconductor chips face-up on a temporary carrier, the plurality of semiconductor chips comprising the conductive studs disposed over active layers of the plurality of semiconductor chips; disposing encapsulant in a single step around four side surfaces of the plurality of semiconductor chips, over the active layers of the plurality of semiconductor chips, and around at least a portion of sidewalls of the conductive studs, wherein a portion of the encapsulant disposed between the plurality of semiconductor chips forms the saw street; and planarizing a surface of the encapsulant and conductive studs. The interior pads may be formed over the plurality of semiconductor chips. The cut tie bar ends may comprise a cross-sectional area in a range of 5-20 micrometers-squared. The land pads, conductive pad finish, or SMS may be electroplated to form a side surface. The QFN, DFN, or SON package may comprise a total height less than 300 micrometers, and may comprise one or more of: multiple independent interior pads; or land pads formed in single row, multiple rows, or as an area array. The conductive layer may be formed comprising reinforcement strips disposed around a periphery of the embedded chip panel. The QFN, DFN, or SON packages may be formed as a System-in-Packages (SiPs) or Multi-Chip-Modules (MCMs) comprising two or more chips in at least one of the QFN, DFN, or SON packages. At least a portion of the conductive layer may be formed using unit-specific patterning to account for actual positions of the plurality of semiconductor chips embedded within the embedded chip panel.

[0007] In another aspect the disclosure relates to a method of making a QFN, DFN, SON package, the method comprising forming an embedded chip panel comprising a plurality of semiconductor chips embedded in encapsulant and separated by saw streets. A conductive layer may be formed over the embedded chip panel, the conductive layer comprising bussing lines, contact pads, traces, and tie bars. The tie bars may extend between and be coupled with the contact pads and the bussing lines, and the bussing lines and tie bars may extend into the saw streets. Vertical conductive elements may be disposed over and coupled with the conductive layer. An encapsulant layer may be disposed over the conductive layer and around the vertical conductive elements such that the vertical conductive elements comprise exposed ends. At least a portion of the tie bars may be exposed from the encapsulant layer. Land pads, a conductive pad finish, or a SMS may be electroplated over the vertical conductive elements by providing a current through the bussing lines and tie bars.

[0008] In an aspect of the disclosure the method may further comprise electroplating a pad surface finish over the

land pads to form a side surface for the land pads. The QFN, DFN, or SON package may comprise one or more of multiple independent interior pads, land pads formed in single row, multiple-rows, LGA, BGA, or as an area array for thermal, power, signal. The embedded chip panel may be singulated through the saw streets, removing a portion of the embedded chip panel to sever the tie bars, and singulating the strips into individual QFN, DFN, or SON packages. The QFN, DEN, or SON packages may be formed as SiPs or MCMs comprising two or more chips in at least one of the QFN, DFN, or SON packages. At least a portion of the conductive layer may be formed using unit-specific patterning to account for actual positions of the plurality of semiconductor chips embedded within the embedded chip panel.

[0009] In yet another aspect, the disclosure relates to a method of making a QFN, DEN, SON package, the method comprising forming an embedded chip panel comprising a plurality of semiconductor chips embedded in encapsulant and separated by saw streets. A conductive layer may be formed over the embedded chip panel and extend into the saw street. A vertical conductive element may be electroplated over the conductive layer by providing a current through the conductive layer. An encapsulant layer may be disposed over the conductive layer, wherein at least a portion of the conductive layer is exposed from the encapsulant layer. Land pads, a conductive pad finish, or a SMS may be electroplated over the vertical conductive element by providing a current through conductive layer.

[0010] In an aspect of the disclosure the method may further comprise an electroplated interior pad disposed over the embedded component. The embedded component may comprise conductive studs coupled to an active layer of the embedded component, and encapsulant may be disposed over the active layer of the embedded component and contact a side surface of the conductive studs. A pad surface finish may be formed over the land pads to form a side surface for the land pads that is not encapsulant defined. The QFN, DEN, or SON package may comprise one or more of multiple independent interior pads, land pads formed in single row, multiple-rows, LGA, BGA, or as an area array for thermal, power, signal. The QFN, DFN, or SON packages may be formed as SiPs or MCMs comprising two or more embedded components in the QFN, DEN, or SON packages. Unit-specific patterning may be used such that a misalignment between the embedded component and the conductive layer is less than a misalignment between the embedded component and a package edge.

[0011] The foregoing and other aspects, features, applications, and advantages will be apparent to those of ordinary skill in the art from the specification, drawings, and the claims. Unless specifically noted, it is intended that the words and phrases in the specification and the claims be given their plain, ordinary, and accustomed meaning to those of ordinary skill in the applicable arts. The inventors are fully aware that they can be their own lexicographer if desired. The inventors expressly elect, as their own lexicographers, to use only the plain and ordinary meaning of terms in the specification and claims unless they clearly state otherwise and then further, expressly set forth the "special" definition of that term and explain how it differs from the plain and ordinary meaning. Absent such clear statements of intent to apply a "special" definition, it is the inventors' intent and desire that the simple, plain and ordinary meaning to the terms be applied to the interpretation of the specification and claims.

[0012] The inventors are also aware of the normal precepts of English grammar. Thus, if a noun, term, or phrase is intended to be further characterized, specified, or narrowed in some way, then such noun, term, or phrase will expressly include additional adjectives, descriptive terms, or other modifiers in accordance with the normal precepts of English grammar. Absent the use of such adjectives, descriptive terms, or modifiers, it is the intent that such nouns, terms, or phrases be given their plain, and ordinary English meaning to those skilled in the applicable arts as set forth above.

[0013] Further, the inventors are fully informed of the standards and application of the special provisions of 35 U.S.C. § 112(f). Thus, the use of the words "function," "means" or "step" in the Detailed Description or Description of the Drawings or claims is not intended to somehow indicate a desire to invoke the special provisions of 35 U.S.C. § 112(f), to define the invention. To the contrary, if the provisions of 35 U.S.C. § 112(f) are sought to be invoked to define the inventions, the claims will specifically and expressly state the exact phrases "means for" or "step for", and will also recite the word "function" (i.e., will state "means for performing the function of [insert function]"), without also reciting in such phrases any structure, material or act in support of the function. Thus, even when the claims recite a "means for performing the function of . . . " or "step for performing the function of . . . ," if the claims also recite any structure, material or acts in support of that means or step, or that perform the recited function, then it is the clear intention of the inventors not to invoke the provisions of 35 U.S.C. § 112(f). Moreover, even if the provisions of 35 U.S.C. § 112(f) are invoked to define the claimed aspects, it is intended that these aspects not be limited only to the specific structure, material or acts that are described in the preferred embodiments, but in addition, include any and all structures, materials or acts that perform the claimed function as described in alternative embodiments or forms of the disclosure, or that are well known present or later-developed, equivalent structures, material or acts for performing the claimed function.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIGS. **1A-1D** show embodiments of conventional, prior art QFN, DEN and SON packages.

[0015] FIG. **1E** illustrates a non-limiting example of a leadframe for mounting ICs or semiconductor chips.

[0016] FIG. **2** illustrates a perspective view of a cross-sectioned portion of a QFN.

[0017] FIG. **3A** illustrates a plan view of a semiconductor wafer or native wafer **10** comprising components disposed thereon.

[0018] FIG. **3B** illustrates disposing semiconductor chips over a temporary carrier.

[0019] FIG. **3C** shows a close-up view of the semiconductor chips taken at section marker **3C** of FIG. **3B**.

[0020] FIG. **3D** illustrates disposing an encapsulant around the semiconductor chips over a temporary carrier.

[0021] FIG. **3E** illustrates a top view of an embedded chip panel, or reconstituted wafer, a portion of which is shown in the cross-sectional profile view of FIG. **3D**.

[0022] FIG. **3**F shows removing a temporary carrier to expose a backside or back surface of the semiconductor chips.

[0023] FIG. **3**G illustrates a profile view of a portion of a semiconductor chip with one or more conductive layers plated over it.

[0024] FIG. **3**H illustrates a top or plan view of a bottom right corner of an embedded chip panel showing portions of nine (9) QFN, DEN, or SON packages being made.

[0025] FIG. **3**I illustrates a top or plan view of the bottom right corner of FIG. **3**H showing a conductive layer.

[0026] FIG. **3**J depicts a semiconductor die embedded within encapsulant as part of a QFN, DEN, or SON package depicting a conductive layer disposed thereon.

[0027] FIG. **3**K depicts a semiconductor die embedded within encapsulant as part of a QFN, DEN, or SON package.

[0028] FIG. **4**A illustrates a perspective view of a QFN, DEN, or SON package after singulation from the embedded chip panel.

[0029] FIG. **4**B shows a cross-sectional side view of the QFN, DFN, or SON package shown in FIG. **4**A.

[0030] FIG. **4**C depicts a cross-sectional side view of a portion of the left side of the QFN, DFN, or SON package shown in FIG. **4**B.

[0031] FIG. **4**D_1 shows a cross-sectional side view of a portion of FIG. **4**C, showing a land pad, LGA pad, or conductive pad with a pad finish or SMS disposed over the land pad.

[0032] FIG. **4**D_2 illustrates the cross-sectional side view of FIG. **4**D_1 during a sawing or singulation process.

[0033] FIG. **4**E_1 shows a cross-sectional side view of a portion of FIG. **4**C showing a vertical conductive element with a pad finish or SMS disposed over the vertical conductive element.

[0034] FIG. **4**E_2 illustrates the cross-sectional side view of FIG. **4**E_1 during a sawing or singulation process.

[0035] FIG. **4**F depicts a cross-sectional side view of a portion of FIG. **4**E after singulation, showing a singulation edge and a corner of a conductive pad comprising a solderable metal system formed according to an embodiment of the present disclosure.

[0036] FIG. **5** illustrates a flowchart for an embodiment of the methods of making a quad flat no-lead (QFN), dual flat no-lead (DFN) or small outline no-lead (SON) package as disclosed herein.

[0037] FIGS. **6**A and **6**B depict cross section views of the conductive layer **68**, additional conductive layer **138**, and plating contact **76**, depicting the formation of LGA pads **142** over contact pads **68***b*.

[0038] FIG. **7** illustrates QFN, DEN and SON packages comprising two semiconductor chips or components.

[0039] FIG. **8** illustrates QFN, DEN and SON packages comprising two or more thermal pads or ground planes.

DETAILED DESCRIPTION

[0040] The present disclosure includes one or more aspects or embodiments in the following description with reference to the figures, in which like numerals represent the same or similar elements. Those skilled in the art will appreciate that the description is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the disclosure as defined by the appended claims and their equivalents as supported by the following disclosure and drawings. In the description, numerous specific details are set forth, such as specific configurations, compositions, and processes, etc., in order to provide a thorough understanding of the disclosure. In other instances, well-known processes and manufacturing techniques have not been described in particular detail in order to not unnecessarily obscure the disclosure. Furthermore, the various embodiments shown in the FIGs. are illustrative representations and are not necessarily drawn to scale.

[0041] The word "exemplary," "example" or various forms thereof are used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as "exemplary" or as an "example" is not necessarily to be construed as preferred or advantageous over other aspects or designs. Furthermore, examples are provided solely for purposes of clarity and understanding and are not meant to limit or restrict the disclosed subject matter or relevant portions of this disclosure in any manner. It is to be appreciated that a myriad of additional or alternate examples of varying scope could have been presented, but have been omitted for purposes of brevity.

[0042] This disclosure relates to a quad flat no-lead (QFN), dual flat no-lead (DFN) or a small-outline no-lead (SON) package without a leadframe and with molded direct contact interconnect build-up structures, and a method of making the same. A QFN, DEN or SON is a small-sized integrated circuit (IC) package that offers small size, low cost, and very good performance. FIG. **1**A illustrates an image of a conventional QFN package **500** with side lengths of 5 mm. Those of ordinary skill in the art are familiar with conventional QFN, DFN and SON package structures.

[0043] No-lead packages such as QFN, DEN and SON packages physically and electrically connect to the surface of printed circuit boards (PCB's) or other substrates using surface mount technology, thus coupling the IC to the PCB or other substrate. In the surface mount technology illustrated in the conventional QFN package **500** of FIG. **1**A and the close-up view in FIG. **1**B, the land pads **502** are exposed on the upper surface **504** of the package **500** and on the side edges **506** of the package **500**. In particular, a portion of the leadframe, called the tie bar **508**, is cut and exposed along the side edges **506** during singulation of the packages. Additionally, when the packages **500** are singulated, such as by sawing or punching, burrs **510** can occur. Singulation by cutting can cause burrs **510** because the land pads extend to the edge **506** of the package **500**, and the saw creates a burr **510** that extends from the land pads **502** caused by the saw heat and rotation as it cuts along the edge of the land pads **502**. Singulation by punching can create burrs due to the need for clearance between the punch and die. In either event, the burring that extends in each of the X-, Y- and Z-planes is a known problem of QFN singulation and requires costly process measures to reduce them or additional processing to remove them.

[0044] FIG. **1**C, and the close-up view of a corner of FIG. **1**C as shown in FIG. **1**D, emphasize that in some conventional packages **520**, a "wettable flank" or wettable vertical surface of a traditional QFN, DEN or SON package extends along a side of the land pad **522**.

[0045] The wettable flank is formed by sawing partially through the package and through the land pad **522** to create the channel **524** or additional edge around a perimeter of the package above the exposed periphery surface **521** of the land pad **522**. This allows the solder fillets (not shown) to be more visible for quality checking after the package is mounted on a board through Automated Optical Inspection (AOI). AOI helps identify that a good electrical connection is formed with the land pad **522** and the solder by seeing when solder is exposed in the channel **524** or at the land pad **522**. In the end, the QFN, DEN and SON packages include exposed copper on the outside, and particularly on the perimeter, surfaces of the packages through the cut land pads **510**, **521** and through the cut leadframe **508**. The exposed copper is prone to oxidation and is undesirable because it makes sidewall solder wetting difficult on the exposed surface **521**.

[0046] QFN, DEN and SON packages are a near chip-scale plastic encapsulated package made with a metal leadframe substrate. FIG. **1**E illustrates a non-limiting example of a leadframe **526** to which ICs or semiconductor chips may be mounted to die pad or flag **527**. The lead tips **528** are separated from the flag **527**, so as to be electrically separated from the flag **527** after singulation.

[0047] A leadframe **526** is the metal structure inside a leadframe-based chip package that creates leads which carry signals and power to and from the chip to the outside through metal conductors leading away from the chip. QFN, DFN and SON packages comprise perimeter land pads **502** and **522** (as seen in FIGS. **1**A-**1**D) on the package edge **506** to provide electrical connections to the PCB, rather than using leads that extend beyond the body of the package. During packaging prior to singulation, the QFN, DEN and SON packages are in strip form, connected to adjacent packages on the leadframe. This also means that they are electrically interconnected through the conductive leadframe. The packages are later isolated or separated, such as by cutting or punching, to form separate packages. In conventional QFN, DEN and SON packages, partial sawing to isolate each unit electrically is required before testing to allow electrical testing of the individual packaged semiconductor chips.

[0048] The present disclosure relates to QFN, DEN and SON packages without a leadframe, and with molded direct contact interconnect build-up structures. An example of a molded direct contact interconnect build-up structure is known under the trademark or tradename MDx™. Molded direct contact interconnect build-up structures (and a method for making and using the same) are discussed in US Provisional Patent 63/347,516, the entirety of which is hereby incorporated herein by reference. Molded direct contact interconnect build-up structures may comprise or provide: (i) large area chip bond pad interconnect to create a very low contact resistance, (ii) removal of capture pads between build-up layers, such as traces, (iii) cost savings by removing polyimide and other polymers from the build-up layers, using mold compound instead, and (iv) facilitate ultra-high-density connections such as 20 micrometer bond pitch and smaller.

[0049] At least some of the above advantages are available at least in part by using unit-specific patterning (such as adaptive patterning (custom design and lithography) and build-up interconnect structures such as a frontside build-up interconnect structure, which is also known under the trademark "Adaptive Patterning," referred to as "AP." Unit-specific patterning: (i) allows for the use high-speed chip attach for semiconductor chips and AP will ensure alignment for high density interconnects with the molded direct contact interconnect build-up structures. Adaptive Patterning may also be used in the herein disclosed processes for manufacturing QFN, DFN and SON packages including the ability to make large area connections which are precisely aligned to chip bond pads for very low contact resistance.

[0050] FIG. **2** illustrates a perspective view of a cross-sectioned portion of a quad flat no-lead (QFN) package **100**, similar to the views shown in FIGS. **3**K and **4**A, which illustrates features that could also be incorporated in a dual flat no-lead (DFN) package **200** or small outline no-lead (SON) package **300** package. The features as shown and described herein may be incorporated into a QFN **100**, a DFN **200** and an SON **300** and as such, where the disclosure refers to specific embodiments, a person of ordinary skill in the art (POSA) would understand that it may apply to any of a QFN **100**, a DFN **200** and an SON **300**. The QFN **100**, DFN **200** or SON **300** package can comprise a semiconductor chip, an embedded chip or embedded component **14** comprising an encapsulant **130**, **130***a* disposed around **4** side surfaces, over active region **20** and contacting front surface **21** of the semiconductor chip **14**. A fan-out, conductive layer **68**, **68***c*, the fan-out layer hereinafter referred to as traces or redistribution layers (RDLs), may be disposed over the semiconductor chip **14** and over the encapsulant **130**, **130***a*. The conductive layer **68***c* may comprise electrically isolated, conductive traces **68***c* and conductive tie bar portions **68***e* coupled with the traces **68***c*, wherein the tie bar portions **68***e* extend from the traces to a package edge **72** where cut tie bar ends **68***f* are exposed with respect to cut encapsulant **156** at edges **72** of the QFN package **100**, DFN package **200**, or SON **300** package. The cut tie bar ends **68***f* may comprise a cross-sectional area in a range of 5-20 micrometers-squared, and a cross-sectional area within this range facilitates singulating the packages using conventional singulation methods such as using a nickel bond blade during sawing or singulation, in contrast to other package structures known in the art which have greater amounts of exposed metal at the package edge. This is further depicted and similarly applies to FIGS. **3**G, **3**K, **4**A-**4**C. A second encapsulant **130**, **130***b* is disposed over the conductive layer **68***c* with openings formed through the encapsulant. Electroplated vertical conductive elements **140** are formed over the conductive layer **68** and contact pads **68***b*. While not depicted for all figures for simplicity, in embodiments according to FIG. **4**D_1, land pads **142** may be electroplated over the vertical conductive elements **140** and may further comprise a pad finish **154** electroplated thereon. In further embodiments according to FIG. **4**E_1, pad finish **154** may be electroplated directly onto vertical conductive elements **140**. The portions of the conductive layer **68** having contact pads **68***b* and contact base **76** make it such that specially formed plating contacts **76** (depicted in FIG. **3**H) on the LGA**1** layer **138** are not needed and contact for electroplating may be made by contacting the conductive layer **68** and contact base **76** to plate both the LGA layer **138** and land pads **142**.

[0051] FIG. **2** further illustrates that the semiconductor chip **14** comprises conductive studs coupled to a front surface **21** over an active layer **20** of the semiconductor chip, and encapsulant disposed over the active layer of the semiconductor chip, which encapsulant contacts a side surface of the conductive studs. Any or all of the QFN, DFN, or SON packages as disclosed herein may be formed using unit specific patterning, where a misalignment between embedded components and conductive layers is less than a misalignment between embedded components and a package edge. A backside laminate **50** may be disposed over backside **18** of chip or component **14** and a backside of encapsulant

130. The backside laminate **50** may be of any suitable dielectric insulating, or passivation layer, that may be the different than encapsulant **130**. In other instances, encapsulant **130** or other material may be present (as illustrated in FIG. **3**K or FIG. **7**), or the backside **18** of the chip or component **14** can be exposed.

[0052] FIG. **3**A illustrates a plan view of a semiconductor wafer or native wafer **10** with a base substrate material **12**, such as, without limitation, silicon, germanium, gallium arsenide, indium phosphide, or silicon carbide, for its semiconducting capability and for structural support. A plurality of semiconductor chip **14** or components can be formed on wafer **10** separated by a non-active, inter-chip wafer area or saw street **16** as described above. The saw street **16** can provide cutting areas to singulate the semiconductor wafer **10** into the individual semiconductor chips **14**.

[0053] Each semiconductor chip **14** may comprise a backside or back surface **18**, an active layer **20** opposite the backside, and a front surface **21**. The active layer **20** contains one or more circuits or discrete components of any kind implemented as one or more active devices, passive devices, conductive layers, and dielectric layers formed within or on the chip **14** and electrically interconnected according to the electrical design and function of the semiconductor chip **14**. For example, the circuit may include, without limitation, one or more transistors, diodes, and other circuit elements formed within active layer to implement analog circuits or digital circuits, such as DSP, ASIC, memory, or other signal processing circuits. The semiconductor chip **14** may also contain passive devices such as inductors, capacitors, and resistors, for RF signal processing, digital power line control or other functions. The semiconductor chip **14** may consist only of conductive routing layers and associated dielectric layers such as for use as a bridge chip between active devices or other electrical function. The semiconductor chip **14** may also be added as one of many chips being added simultaneously on a carrier. The semiconductor chip **14** may also be only a dummy substrate with no electrical function, but rather act merely as a structural element, and in such instance may be referred to as an embedded device **14**. Embedded device **14** may further comprise one or more of a passive device, dummy chip, bridge chip, and IPD sensor. In some instances, the embedded device **14** may comprise connections on both sides of the chip **14**. The principles and structures taught in relation to this disclosure are applicable to known existing technologies that are compatible with the QFN, DEN or SON packages disclosed without a leadframe and using direct contact interconnect build-up.

[0054] FIG. **3**B illustrates disposing semiconductor chips **14** face-up over a temporary carrier **120**, the semiconductor chips each comprising conductive studs **125** over the active layer **20** of the semiconductor chips **14**. Face-up as used herein means where the conductive studs **125**, active layer **20** and front surface **21** of the semiconductor chip **14** are disposed away from temporary carrier **120** such that backside **18** contacts temporary carrier **120**. Face-down means where the conductive studs **125**, active layer **20** and front surface **21** of the semiconductor chip **14** are disposed towards temporary carrier **120**.

[0055] In other instances, semiconductor chips **14** may be disposed face-down over the temporary carrier **120**, and then subsequently have encapsulant **130**, **130***a* disposed around the chips **14**, similar to what is described with respect to FIG. **3**D. In instances where the semiconductor chips **14** are mounted face-down, the conductive studs **125** may be formed before or after the encapsulant **130**, **130***a* is disposed around the semiconductor chips **14**. In some instances, conductive studs **125** will be formed before encapsulant **130**, **130***a* is applied, such as by molding, and the encapsulant **130**, **130***a* may be vacuum drawn around the conductive studs **125**. In other instances, the semiconductor chips **14** may be placed face down on the temporary carrier **120** and molded without conductive studs **125** being present; after which, the temporary carrier **120** is removed to expose chip bond pads over which conductive studs **125** may be optionally formed then overmolded, to yield a structure like that illustrated in FIG. **3**F.

[0056] The conductive stud **125** is a conductive interconnect structure that has generally vertical sides and is typically wider than it is tall, built-up on a substrate, such as over an active layer of a chip, polyimide or mold compound. The conductive stud **125**, though typically formed of the same materials as a pillar or post would be formed, is different than a pillar or post which each has a height greater than its width. The conductive stud **125**, though it commonly is formed in a cylindrical shape, may be formed in any polygonal or other shape and size. Another use for a conductive stud **125** is as a dummy thermal conductive stud **125***a* that is not electrically coupled to an active electrical circuit but is instead thermally coupled to a heat source of an active device to dissipate the heat to another structure, such as to a pad on a surface of the package. The generally vertical sides of a conductive stud **125** are different from the sides shape that exists for a solder ball or a squished out solder ball that has generally rounded sides, because the generally vertical nature of the conductive stud **125** comes from imperfections in being formed in a structure that has been previously developed or etched, such as within openings in a photoresist layer. Developing or etching does not generally perfectly or uniformly remove the photoresist within the openings, and therefore forms imperfect, generally vertical openings for deposition of the conductive stud **125**. Generally vertical includes perfectly vertical and imperfectly vertical sides. The conductive stud **125** is not a wire bond or solder.

[0057] In some instances, the semiconductor chips **14** will have a thickness (shown in the vertical direction, bottom to top, of the page) of between about 25 μm to about 150 μm for thin ground wafers, or about 100 μm to about 800 μm for thick ground wafers. In some instances, the temporary carrier **120** may be a metal carrier, a silicon carrier, a glass carrier, or a carrier made of other suitable material used for the molding or encapsulating process, and then be removed after the encapsulant **130**, such as mold compound, filled epoxy film such as ABF, or other dielectric such as polyimide has been placed, cured, or both, such that the encapsulant **130** provides structural support and the temporary carrier **120** is no longer needed for processing. The semiconductor chips **14** may be placed adjacent one another, such as in a side-by-side arrangement, so that multiple chips **14** may be formed at a re-constituted wafer or panel level **30** and processed through various fabrication steps, before being singulated into individual QFN, DEN or SON packages. As such, multiple chips **14** may also be processed together at a same time over the temporary carrier **120**, which will be understood by a person of ordinary skill in the art (POSA), even when a close-up view of just portions of the semiconductor chips **14** are shown.

[0058] FIG. **3**C illustrates a close-up view of the semiconductor chips **14** taken at section marker **3**C of FIG. **3**B, emphasizing the conductive studs **125** (e.g. may be formed of copper) having sidewalls **126** may be formed over the active layer **20** and over front surface **21**, and aligned on the semiconductor chips **14**. Although it is not required for every embodiment, the conductive studs **125** of the particular embodiments illustrated herein are shown about a perimeter of the semiconductor chips **14** for the particular implementation in which they will be used. In other embodiments, such as for power and ground or thermal management, conductive studs **125** and thermal studs **125***a* may further be disposed within an interior of the semiconductor chips **14**. An optional interface layer **122**, such as double-sided tape, film or deposited material, may be used beneath the semiconductor chips **14** to temporarily hold them to the temporary carrier **120** during processing.

[0059] FIG. **3**D, continuing from FIG. **3**C, illustrates forming an embedded chip panel **30** comprising a plurality of semiconductor chips embedded in encapsulant and separated by saw streets. The embedded chip panel **30** is formed by disposing an encapsulant **130**, **130***a* around the semiconductor chips **14** face-up over a temporary carrier **120** around four side surfaces of the semiconductor chip **14**, over the active layer **20** of the semiconductor chips **14**, and around the conductive studs **125**, and contacting sidewalls **126** of the conductive studs **125**. As used herein, over, on or around may mean in direct contact with, or with other intervening layers, such as polymer or polyimide layers disposed between the chips **14** and the encapsulant **130**. The conductive studs **125** formed over the active layer **20** of the semiconductor chips **14** may be in contact with, surrounded by, partially encircled by, or encapsulated or molded with a single encapsulant, polyimide or mold compound at a single step such that the same encapsulant, polyimide or mold compound **130** is disposed around the semiconductor chips **14**. The encapsulant **130** can be deposited around the plurality of semiconductor chips **14** using a paste printing, compression molding, transfer molding, liquid encapsulation, dispensing, lamination, vacuum lamination, spin coating, slit or slot die coating, or other suitable application. The encapsulant **130** comprises an organic material, a mold compound, a polyimide, a composite material, such as epoxy resin with filler, such as ABF or epoxy acrylate with filler, and is a material suitable for planarizing, such as through chemical mechanical planarizing (CMP) or grinding. As such, in some instances the encapsulant **130** will not be a polymer material, such as an un-filled polyimide, that does not perform well in a grinding operation, and may gum-up a grinding wheel. FIG. **3**F illustrates the exposed planar surface **132** with the interconnect structures **125** exposed.

[0060] FIG. **3**E illustrates a top view of the embedded chip panel, or reconstituted wafer **30**, a portion of which is shown in the cross-sectional profile view of FIG. **3**D. FIG. **3**E illustrates a temporary carrier, a reusable carrier, a sacrificial carrier, or any suitable carrier **120**, made of metal, glass, silicon, mold compound, or other suitable material, with a release layer. The carrier **120** may comprise a form factor or footprint of a wafer (circular footprint), a panel (square or rectangle), or of any suitable shape and may comprise a diameter or width of 200-600 mm, such as 300 mm, or of any other suitable size. The embedded chip panel **30** comprises semiconductor chips **14** disposed in encapsulant **130**.

[0061] FIG. **3**F, continuing from FIGS. **3**D and **3**E illustrates that after molding, the temporary carrier **120** may be removed, and a backside or back surface **18** opposite front surface **21** of the semiconductor chips **14** may be exposed from the encapsulant **130**, **130***a*. Alternatively, a backside laminate, encapsulant, die attach film (DAF) or other material may be disposed over the backside of the semiconductor chips **14**. Thus, in some instances the backside laminate (including polyimide or mold compound) may be more than temporary and may become part of the final product, or may be removed at a later process step, such as at a grinding or polishing step. Different processing steps may be included to result in the backside **18** of the semiconductor chip **14** being exposed with respect to, or covered by, encapsulant or other material.

[0062] FIG. **3**F further illustrates planarizing or grinding the encapsulant **130**, **130***a* over the active layer **20** or front surface **21** to expose the conductive studs **125**, which may occur before or after removing the temporary carrier **120**. Planarizing the encapsulant over or front surface **21** of the semiconductor chips **14** to create a planar surface **132** further exposes ends or planarized ends **128** of the conductive studs **125**, also forming a planarized encapsulant surface **132***a*. The planarizing or grinding of the encapsulant **130**, **130***a* produces a flatness of within a range of about 0.5-5micrometers and a total roughness height from peak to valley measured over a 1 millimeter (mm) length of between 5 and 500 nanometers (nm). While conventional encapsulant grinding might be done with less flatness, greater accuracy and precision can be obtained by using integrated sensors such as laser, acoustic, or other non-contact methods to control the grinding resulting in better flatness. In some instances, the conductive studs **125** may be formed with a height of less than or equal to about 50 micrometers (μm) or less than or equal to about 250 μm, and then be ground down to a height of less than its original height, such as, in a particular embodiment, less than or equal to about 4 μm or 1 μm. As used herein, "about" or "substantially" means a percent difference less than or equal to 50% difference, 40% difference, 30% difference, 20% difference, 10% difference, or 5% difference.

[0063] FIG. **3**G illustrates a profile view of a portion of semiconductor chip **14**, similar to the view of FIG. **3**F, and forming a conductive layer **68** over the embedded chip panel **30** (the conductive layer comprising one or more first formed seed, barrier, and adhesion layers) plated over the flat or planar surface **132** of encapsulant **130**, **130***a,* over the planarized encapsulant surface **132***a* and over planarized ends **128** of the conductive studs **125** as seen in FIG. **3**F. A blanket seed layer (which may further comprise barrier and/or adhesion layers) may be formed over the planar surface **132** of first encapsulant **130**, **130***a,* followed by photoresist deposition, cure and patterning (not shown). Thereafter the conductive layer **68** may be electrodeposited, followed by removal of the photoresist by dry or wet stripping processes as known in the art. The blanket seed layer may be chemically etched for removal. The conductive layer **68** may comprise traces **68***c,* bussing lines **68***a,* tie bars **68***e,* contact pads **68***b,* contact base **74** (as seen in FIG. **3**H) and other features as discussed in more detail with respect with FIGS. **3**H, **3**I and **3**J, formed over one or more semiconductor chips **14**. The traces **68***c* may extend beyond an edge of the semiconductor chips **14** and coupled with conductive studs **125** coupled to the plurality of semiconductor chips **14** and bussing lines **68***a* may be formed so as to pass over one or more saw streets edges or package edges **72**.

[0064] After a second photoresist deposition, cure and patterning as previously discussed, an additional conductive layer or land grid array (LGA**1**) layer **138** (shown in FIG. **3**H) may subsequently be electroplated to form vertical conductive elements **140** and in some instances to form pads, thermal pads, center exposed interior pads or ground pads **138**, **138***a* over the conductive layer **68**. The additional conductive LGA**1** layer **138** may be electroplated by providing current through the bussing lines **68***a* and tie bars **68***e* of the conductive layer **68** (RDL layer), which may extend to an edge **72** of the package. Thus, the land pads **142** and interior pads **138***a* may be formed as part of additional conductive layer **138** by electroplating without the use of a barrier layer or adhesion layer or seed layer. The second photoresist layer may be removed as aforementioned and as known in the art. In further embodiments, a photosensitive polyimide (PSPI) in lieu of the first or second photoresist layer may be used.

[0065] With respect to the seed layer **131** (shown in FIGS. **4**D_1-**4**F), and the subsequently formed conductive layer or RDL layer **68**, FIG. **3**G shows an electrically conductive layer **68** is formed over the embedded chip or semiconductor chip **14**. The one or more seed **131**, barrier, or adhesion layer and conductive layer **68** can be one or more layers of Al, Cu, Sn, Ni, Au, Ag, Ti—Cu, TiW—Cu, TiN—Cu, Ta—Cu, TaN—Cu or a coupling agent like Cu. The seed layer **131** and conductive layer **68** may be deposited by sputtering, by electroless plating, by electroplating, by depositing laminated Cu foil combined with electroless plating, or other suitable process. In an embodiment, the seed layer **131** and the conductive layer **68** is flat or planar. The seed layer **131** may cover an entirety of an upper surface of the embedded chip panel **30** and form a plating buss for subsequent plating of the conductive layer. A patterning layer (not shown; such as a patterned photoresist layer, discussed below) can direct where the conductive layer is formed. Thus, the seed layer and the conductive layer can cover both the top surface or an area over semiconductor die **14** as well as a peripheral area disposed outside a footprint of the semiconductor die, such as gaps and saw streets.

[0066] A patterning layer, such as photoresist, may be conformally applied over the embedded chip panel **30** and be patterned to facilitate the formation of the conductive layer **68**. In an embodiment, the patterning layer is a dry film resist layer. The patterning layer may be patterned and a portion of the patterning layer may be removed by developing, laser direct imaging (LDI) and developing, etching, laser drilling, or other suitable process to form openings completely through patterning layer and to expose the conductive seed layer **131** for subsequent plating or formation of the conductive layer **68**. Patterning layer may be patterned to facilitate the formation of traces **68***c,* bussing lines **68***a,* tie bars **68***e,* and reinforcement strips **74**. The patterning layer may also be patterned to facilitate the formation of contact pads **68***b* and traces or redistribution layers (RDLs) **68***c* as part of a fan-out interconnect structure for semiconductor chips **14** that are coupled or connected to the bussing structures **68***a* when plated as shown, e.g., in FIGS. **3**H-**3**J, as described in greater detail following.

[0067] FIG. **3**H illustrates a top or plan view of a bottom right corner of an embedded chip panel **30** showing portions of nine (9) QFN, DEN, or SON packages being made. The conductive layer (or RDL layer) **68** shown in FIG. **3**H may be formed over the embedded chip panel **30**. The conductive layer **68** comprises traces **68***c,* bussing lines **68***a,* tie bars **68***e,* and a contact base or reinforcement strip **74**, which may comprise part of conductive layer **68**, for subsequently formed plating contacts **76**, which are part of the LGA**1** layer **138**. In the final plating configuration, contact base or reinforcement strip **74** may extend under plating contacts **76** for electrical continuity and because plating contacts **76** tie into the plating busses **68***a* formed in **68**, plating contacts **76** can subsequently be used to electroplate the pad finish materials or SMS **154**. The traces or RDLs **68***c* are formed over the plurality of semiconductor chips **14** and coupled with conductive studs **125** coupled to the plurality of semiconductor chips **14**. The bussing lines **68***a* are formed over, and extend along or through, the saw street edges or package edges **72** (shown in FIGS. **31** and **3**J, following) between the semiconductor chips **14**. The tie bars **68***e* extend between, and are coupled with, the traces **68***c* and the bussing lines **68***a*. The contact base or reinforcement strip **74** for subsequently formed plating contacts **76** can also extend outwards around a perimeter of the embedded panel or reconstituted wafer **30** to help provide mechanical strength and provide a connection to the subsequently formed plating contacts **76** as part of the LGA**1** layer **138**.

[0068] The conductive layer **68** can also be formed using unit-specific patterning to align the conductive layer **68** with a true position of a plurality of semiconductor die **14**. When positions of semiconductor die **14** and conductive studs **125** shift from nominal positions such as during placement and encapsulation of the semiconductor die for formation of the embedded chip panel **30**, the true or actual positions of the semiconductor die **14** may not sufficiently align with the nominal design of the conductive layer **68**, including RDL and trace positions, for the structure to provide desired reliability for package interconnections given desired routing densities and pitch tolerances. When shifts in the positions of semiconductor die **14** are small, no adjustments to the positions of the conductive layer **68** may be required to properly align stud bonding portions **68***i* as part of the conductive traces **68**, **68***c* with the conductive studs **125**. However, when changes in the positions of semiconductor die **14** are such that the nominal position does not provide adequate alignment with, and exposure to, the conductive studs **125**, then adjustments to the position of the conductive layer **68** can be made by unit-specific patterning (or "adaptive patterning™" as described in greater detail in U.S. patent application Ser. No. 13/891,006, issued as U.S. Pat. No. 9,196,509, titled "Adaptive Patterning for Panelized Packaging" filed May 9, 2013, the disclosure of which is hereby incorporated herein by reference. Unit-specific patterning can optionally adjust the position of features of the conductive layer (including traces, bussing lines, and tie bars) for each semiconductor die **14** individually, or can adjust positions for a number of semiconductor die **14** simultaneously. The position, alignment, or position and alignment of features of semiconductor die **14** can be adjusted by an x-y translation or by rotation of an angle **0** with respect to their nominal positions or with respect to a point of reference or fiducial on the embedded chip panel **30**.

[0069] An additional conductive layer (LGA**1**) layer **138** may be formed over the conductive layer **68**, as shown in FIG. **3**H. The additional, LGA**1** layer **138** may be formed by electroplating land pads **142**, thermal pads or ground planes **138***a,* and plating contacts **76** within the additional conductive LGA**1** layer **138** by providing a current through the conductive layer **68** (including through the bussing lines and tie bars), by which the land pads **142**, thermal pads or ground planes **138***a,* and plating contacts **76** are electroplated without a temporary photoresist layer, without a seed layer or an adhesion-barrier layer, and within the openings formed through the second encapsulant **130**, **130***b*. The plating contacts **76** may be disposed at a periphery of the reconstituted wafer or panel **30** and separated from the plurality of semiconductor chips **14** by a portion of the saw streets **72** and the contact base **74** ("reinforcement strip"). In some embodiments, the conductive layer **68** may comprise reinforcement strips or contact base **74** formed around a periphery of the embedded chip panel **30**.

[0070] The LGA**1** layer **138** can be one or more layers of Cu, Sn, Ni, Au, Ag, or other suitable electrically conductive material. The deposition of the LGA**1** layer **138** may be formed as part of an additive process on a molded panel **30** using a plating process (such as electrolytic plating or electroless plating) as described. In some other instances, another additive process may be used, such as PVD, CVD, or other suitable process. In an embodiment, the conductive LGA**1** layer **138** is formed over the seed layer by an electro-plating process that uses the seed layer as a plating surface. The LGA**1** layer **138** or land pads **142** may also be formed using unit-specific patterning to align the conductive layer with a true position of a plurality of semiconductor die (as described in greater detail below). LGA**1** layer **138** comprises a thickness in a range of 10-35 micrometers (μm), or a thickness greater than 20 μm, or a thickness greater than 25 μm. Land pads **142** provide mechanical and electrical interconnection, such as with a subsequently formed tin layer, bumps, or package interconnect structures that provide for the transmission of electrical signals between semiconductor die **14** and points external to the QFN, DFN, or SON packages. The interior pads **138**, **138***a* can provide multiple benefits, including a thermal conduction path for cooling of semiconductor die **14** (i.e. heat transfer and heat dissipation), as well as provide an electrical ground or power bus for the package. The plating contacts **76** may be electroplated over the conductive layer **68** by providing a current through the bussing lines and tie bars, wherein the plating contacts **76** are electroplated without a photoresist layer and within the openings formed through the encapsulant, wherein the plating contacts **76** comprise a thickness greater than or equal to 25 micrometers of copper. The plating contacts **76** (together with the reinforcement strips **74** below them) provide a reinforced structural feature and conductive contact points for subsequent plating, such as with a detachable conductive ring that may be disposed around the embedded chip panel **30** for subsequent plating. Subsequent plating comprises electroplating as one or more layers of matte tin, Ni—Pd—Au, Ni—Au, Au, Ag, and solder over the land pads **142**, thermal pads and/or ground pads. In some embodiments, subsequent plating further comprises electroplating a pad surface finish **154** by providing a current through the bussing lines and tie bars, over the land pads **142** and/or over the interior pads, thermal pads or ground pads **138**, **138***a* as one or more layers of matte tin, Ni—Pd—Au, Ni—Au, Au, Ag, and solder.

[0071] FIG. **3**I illustrates a top or plan view of a bottom right corner as shown within the dashed box of FIG. **3**H, and without the LGA**1** layer **138** formed over the conductive layer or RDL layer **68**. The conductive layer **68** comprises traces **68***c,* comprising stud bonding portions **68***i,* bussing lines **68***a,* tie bars **68***e,* and reinforcement strips or contact base **74** disposed between and around the traces **68***c*. The conductive layer **68**, or portions thereof, may be formed using adaptive patterning.

[0072] FIG. **3**J shows a simplified schematic plan view (with respect to FIG. **31**) of the conductive layer **68** for one package. FIG. **3**J shows semiconductor die **14** embedded within encapsulant **130**, **130***a* as part of a QFN, DEN, or SON packages having a package outline **70** defined by the saw street edges **72**. Conductive layer **68** is shown formed into a number of features **68***a*-**68***g* and **68***h* as part of the QFN, DFN, or SON package. Conductive layer **68** includes bussing lines or a bussing structure **68***a* that connects various features of conductive layer **68** and facilitates plating of additional conductive material for the formation of input/output (I/O) interconnect structures. A portion of bussing lines **68***a* are formed in saw streets **72**, that can surround a periphery of the QFN, DFN, or SON package, and are subsequently removed during singulation of individual QFN **100**, DEN **200**, or SON **300** packages through the saw streets **72**. The conductive layer **68** comprises traces **68***c,* bussing lines **68***a,* tie bars **68***e*. Disposed between the tie bars **68***e* and traces **68***c* are contact pads **68***b* as part of conductive layer **68**, onto which LGA pads **142** may subsequently be formed as further described following. Similarly, additional layers, such as one or more ground planes, non-electrically conductive fill material which is electrically isolated, electrically conductive fill material providing electrical benefit as a ground plane, power plane, or thermal pad and similar features may be formed over the center exposed interior pad or ground pad **68***d.* Tie bars **68***e* may extend between and coupled with the traces **68***c,* bussing lines **68***a,* and plating contacts **68***b*. The method includes singulating the embedded chip panel **30** through the saw streets **72** using saw **160**. In some aspects of the method, the embedded chip panel **30** may be singulated using saw **160** having a nickel bond blade (shown in FIGS. **4**D_2 and **4**E_2) to form a plurality of QFN, DFN, or SON packages with exposed ends **68***f* of the tie bars **68***e* exposed at a periphery of the QFN, DFN, or SON packages. After singulation, the tie bars **68***e* may comprise cut tie bar ends **68***f,* where the cut tie bar ends **68***f* may comprise a height in a range of 1-15 micrometers and a width in a range of 1-15 micrometers, and a cross-sectional area in a range of 5-20 micrometers-squared. In some embodiments, after singulation the tie bars **68***e* may comprise a length less than or equal to 10 micrometers.

[0073] Conductive layer **68** also includes contact pads **68***b* that provide a location for the subsequent formation of vertical conductive elements **140**, LGA pads **142** or package I/O interconnects. Contact pads **68***b* are mechanically and electrically coupled or connected to conductive studs **125** by traces or RDLs **68***c*. In some instances, metal fill may be disposed around or adjacent traces or RDLs **68***c,* as shown, e.g., in FIG. **3**I. Traces or RDLs, **68***c,* may be fanned-out over each of the plurality of semiconductor chips (as shown by FIG. **3**I) from conductive studs **125** (which are coupled to the plurality of semiconductor chips **14**) which are coupled to a stud bonding portion **68***i* of conductive layer **68**, and extending to contact pads **68***b*. Contact pads **68***b* can be disposed side-by-side a first distance **68***h* from package outline **70**. Alternatively, contact pads **68***b* are offset in multiple rows such that a first row of contact pads is disposed the first distance **68***h* from package outline **70**, and a second row of contact pads alternating with the first row is disposed a second distance from the package outline. In some embodiments, the first distance **68***h* may comprise a distance in a range of 5-100 μm, or 10-50 μm.

[0074] A portion of conductive layer **68** is optionally formed as part of a center exposed interior pad, flag or thermal pad **68***d*. The interior pad **68***d* may be a ground pad, a power pad (comprising a positive voltage, a thermal pad that is an electrically unconnected structure that is physically present as a thermal pad. The interior pad **68***d* may be coupled or connected to ground, power, or a thermally transmissive area of (or in) the semiconductor die **14**, such as through one or more conductive studs **125** or thermal studs **125***a*. In an embodiment and as illustrated in FIG. **3**K, an interior or ground pad **68***d*, either alone or with additional conductive material (LGA**1**) **138** formed as center exposed pad, thermal pad, ground pad **138***a*, may be exposed at a bottom surface or at a periphery of the completed semiconductor package and may be disposed entirely within a footprint **14***f*, of semiconductor die **14**. Alternatively, interior or ground pad **138***a* may be disposed partially within a footprint **14***f*, of semiconductor die **14**, and in further embodiments, a footprint **138***a* of interior or ground pad **138***a* may extend outside footprint **14***f* of semiconductor die **14** as shown in the embodiment of FIG. **4**B, depicting interior pad, thermal or ground pad **144** having footprint **144***g* extending beyond footprint **14***f*, of semiconductor die **14**. Thus, conductive layer **68** can be initially formed such that bussing lines **68***a*, contact pads **68***b*, RDLs **68***c*, ground pad **68***d*, and tie bars **68***e* are electrically common before the removal of bussing lines **68***a*, and the cutting of tie bars **68***e*, such as by singulation. After the removal of bussing lines **68***a*, and the cutting of tie bars **68***e*, as shown and described below with respect to FIG. **4**A, RDLs or traces **68***c* are not electrically common and are electrically discrete.

[0075] A second encapsulant **130**, **130***b* may be formed over the conductive layer **68** and over the additional conductive layer **138**. In some instances, the second encapsulant **130**, **130***b* formed over the conductive layers may be similar or identical to the encapsulant layer or mold compound **130**, **130***a* formed around the semiconductor chips **14**. In other instances, the second encapsulant **130**, **130***b* formed over the conductive layer **68** may be a photo-sensitive material (e.g., photosensitive polyimide (PSPI)). In yet other instances, the second encapsulant **130**, **130***b* layer may be patterned and chemically etched.

[0076] FIG. **3**K depicts an embodiment of a QFN **100**, DFN **200**, or SON **300** package after singulation from an embedded chip panel **30**. Shown are conductive studs **125** formed over a front surface **21** of semiconductor chips **14** and coupled with traces **68**, **68***c*. Further illustrated are thermal studs **125***a*, which may be formed over a front surface **21** of the plurality of semiconductor chips **14** and coupled with center exposed interior pad, thermal pad, or ground pad **68***d* to provide thermal dissipation or electrical grounding from the semiconductor chips **14** to the pad **68***d* and any external devices. In the embodiment of FIG. **3**K, the center exposed interior pad, thermal pad, or ground pad **68***d* may be disposed entirely within a footprint **14***f* of semiconductor die **14**. Further illustrated is first distance **68***h* from package edge **70**. FIG. **3**K illustrates like elements as shown and described for FIG. **2**. In some instances, the package **100**, **20**, **300** of FIG. **3**K may comprise a backside laminate **50**, similar to, or as shown in FIG. **2**.

[0077] FIG. **4**A illustrates a perspective view of a QFN, DEN, or SON package after singulation from the embedded chip panel **30**. After singulation, the conductive layer **68** comprises electrically isolated traces **68***c* and tie bar portions **68***e* coupled with the traces **68***c*. Tie bar portions **68***e* extend from the traces **68***c* to a package edge **72** where cut tie bar ends **68***f* are exposed with respect to cut encapsulant **156** at edges **72** of the QFN, DEN, or SON package. A pad surface finish layer or solderable metal system **154** may be electroplated over one or more of the land pads **142** and over thermal pad, heat sink, or exposed heatsink **144**. The land pads **142** comprise a number of side surfaces **142***a* that are not encapsulant defined and are offset by the first distance **68***h* from package or singulation edge **72** of the QFN, DFN, or SON package. The side surface **142***a* may extend around a perimeter of the land pads **142**, increasing solder joint strength when the QFN, DFN, or SON package is mounted to a circuit board or other external component. While FIG. **4**A illustrates the land pads and center exposed pads protruding a relatively large distance from the upper surface of the encapsulant, the land pads and center exposed pads may be flush or equal in height with the upper surface of the encapsulant, or only slightly offset above or below the upper surface of the encapsulant.

[0078] While the QFN, DEN, or SON package comprises exposed copper in the final package with the cut tie bar ends **68***f*, an amount of exposed conductive material (e.g., copper) will be less than amount of exposed conductive material (e.g., copper) than with a conventional QFN, DFN, or SON package that is built using a leadframe. Creating the tie bar connection **68***e* allows for a suitable pad finish or bonding material (e.g., tin or a SMS) to be disposed on the LGA or land pad (as well as the center exposed pad) by electrolytic plating, which is less expensive than using immersion plating, which is much more expensive. Additionally, by having a reduced amount of copper being sawn through during singulation (as with respect to a package comprising a conventional leadframe), better sawability and singulation is provided with the current design. The improved sawability and singulation allows for use of a nickel bond blade (a same saw blade as what is used when no copper is present in saw streets), which is an order of magnitude better for the cost of sawing and singulation. Additionally, singulation can occur with a single sawing step, and not use a two-step sawing process, such as when a first larger cut is used to remove copper and form a step, followed by a second cutting that goes through the first cutting to singulate the packages. Forming the package without a step (according to the present design) provides a connection that is better than conventional connections formed with a step. Instead, forming the package without a step (according to the present design) provides improved solderability and a connection that is equal to or better than the connections or solderability of a package formed using a conventional two-step sawing process.

[0079] FIG. **4**B illustrates cross-sectional side view of the QFN, DFN, or SON package shown above in FIG. **4**A. In some embodiments and as depicted in FIG. **4**B, the QFN, DFN, or SON package may comprise an interior pad, center exposed interior pad, thermal pad or ground pad **144** having a footprint **144***g* extending beyond a footprint **14***f* of the semiconductor chip **14**. In some instances, the QFN, DFN, or SON packages, while having many of the same or similar features as in **3**K for example, may comprise more than one component or semiconductor chip **14**, **14***a* and **14***b,* as shown in FIG. **7**. This allows for the combination of components or chips **14** having different functionality. Further illustrated in the embodiment of FIG. **4**B is where a backside **18** of the semiconductor chip **14** is exposed for thermal management. In addition to the other advantages described above, the present design allows for strip test compatibility, with a reduced isolation cost. For example, isolation and testing may be accomplished as part of process that might comprise one or more of the following features: (i) a laser or a saw may be used to remove a portion of the reconstituted wafer **30** to sever the bussing lines **68***a* and the tie bars **68***e* among the various packages to electrically isolate features before electrical testing; (ii) some or more of the reconstituted wafer **30** may be removed so as to separate the reconstituted wafer into separate strips comprising a plurality of QFN, DFN, or SON packages for strip testing; and (iii) at some point, including after strip testing, the separate strips may be singulated into individual QFN, DFN, or SON packages.

[0080] In other instances, one or more of more of the following features may be incorporated: (i) remove more of the embedded chip panel to separate the embedded chip panel into separate strips comprising a plurality of QFN, DFN, or SON packages for strip testing; (ii) plate an SMS; (iii) remove with a laser or a saw a portion of the embedded chip panel to sever the bussing lines and the tie bars; and (iv) after strip testing, singulating the strips into individual QFN, DFN, or SON packages.

[0081] FIG. **4**C illustrates an enlarged close-up cross-sectional side view of the left side of the QFN, DFN, or SON package shown above in FIG. **4**B. FIG. **4**C further illustrates that the QFN, DEN, or SON package comprises a total height or thickness (T) less than or equal to 300 micrometers (μm) which is less than a conventional thin QFNs or Ultra-Thin QFN (UQFN) that will be specified as nominally comprising a thickness of 0.55 millimeters (mm) or 550 micrometers (μm). The QFN, DEN, or SON package of FIG. **4**C may further comprise a total height T less than or equal to 200 μm, or less than or equal to 100 μm. The process of record (POR) or method for forming the QFN, DFN, or SON may comprise forming or disposing the encapsulant or EMC **130** over or around the conductive studs **125**, planarizing the encapsulant **130** and ends **128** of the conductive studs **125**, forming or plating-up the conductive layer or RDL **68**, form or plate one or more of the vertical conductive elements **140**, the land pads or LGA **142**, and interior pad(s) **138***a* as part of additional conductive layer **138**, disposing an additional encapsulant layer over the conductive layers or disposing a photosensitive material like photosensitive polyimide (PSPI) over the conductive layers, expose and develop the photosensitive material, and then plate the pad finish (solderable finish or SMS) **154** by providing a current through the bussing lines **68***a,* plating contact **76**, and tie bars **68***e*. The QFN, DFN, or SON may then be singulated, as shown and discussed below with respect to FIG. **4**E and **4**F. In further embodiments, the QFN, DFN, or SON packages may be formed as System-in-Packages (SiPs) or Multi-Chip-Modules (MCMs) comprising two or more chips in at least one of the QFN, DEN, or SON packages. As shown in FIG. **4**C, additional conductive layer or land grid array (LGA**1**) layer **138** may form vertical conductive element **140** and thermal pad, or ground pad **138***a*. The vertical conductive element **140** may be a conductive interconnect structure that has generally vertical sides and is typically wider than it is tall, built-up on a substrate, such as over an active surface of a chip, polyimide or mold compound. The vertical conductive element **140**, though typically formed of the same materials as a conductive stud **125** would be formed, is different than a conductive stud **125** which typically has a height equal to or greater than its width. The vertical conductive element **140** may be disposed between, and coupled to, conductive layers **68** and **138**, while a conductive stud **125** is formed coupled to a semiconductor chip **14**. The vertical conductive element **140**, though it commonly is formed in a cylindrical shape, may be formed in any polygonal or other shape and size. The vertical conductive element **140** may be formed by electroplating. The generally vertical sides of the vertical conductive element **140** are different from the sides shape that exists for a solder ball or a squished out solder ball that has generally rounded sides, because the generally vertical nature of the vertical conductive element **140** comes from imperfections in being formed in a structure that has been previously developed or etched, such as within openings in a photoresist layer. Developing or etching does not generally perfectly or uniformly remove the photoresist within the openings, and therefore forms imperfect, generally vertical openings for deposition of the vertical conductive element **140**. Generally vertical includes perfectly vertical and imperfectly vertical sides. The vertical conductive element **140** is not a wire bond or solder. In some embodiments, thermal pad, or ground pad **138***a* may comprise one or more ground planes, non electrically conductive fill material which is electrically isolated, or electrically conductive fill material providing electrical benefit as a floating ground plane.

[0082] In further instances of the method as disclosed herein, a molding process may be performed to form second encapsulant **130**, **130***b,* over conductive layer **68** and prior to forming additional conductive layer **138**, followed by a laser process to form openings for forming the electroplated land pads **142**, and land pads **142** as part of additional conductive layer **138**, over the conductive layer **68** and contact pads **68***b* through laser-defined openings in the second encapsulant **130**, **130***b*. In some instances, a PSPI may be used as the second encapsulant **130**, **130***b* layer and may be patterned and chemically developed. In yet other instances, the second encapsulant **130**, **130***b* layer may be etched by a laser. The openings in the second encapsulant **130**, **130***b* may expose portions of the conductive layer **68** having contact pads **68***b* for electrodeposition of the LGA pads **142** and interior pad **138***a*.

[0083] FIG. **4**D_1, continuing from FIG. **4**C, is an embodiment of a cross-sectional side view of an enlarged portion of FIG. **4**C indicated by detail mark **4**D. As shown in FIG. **4**D_1, seed layer **131** may be formed over encapsulant **130**, **130***a,* (such as over planarized encapsulant surface **132***a* of FIG. **3**F) and conductive layer **68**, formed comprising the features as disclosed for FIG. **3**J, may be formed over seed layer **131**. Further shown is forming vertical conductive elements **140** disposed over and coupled with the conductive layer **68**. FIG. **4**D_1 further illustrates the second encapsulant **130**, **130***b,* or PSPI disposed over the conductive layer **68** and around and contacting vertical conductive elements **140**. Land pads or conductive pads **142** with the pad finish or SMS **154** disposed over the land pad **142** may be disposed over the second encapsulant **130**, **130*b*** and coupled to vertical conductive elements **140**. According to further embodiments, an additional seed layer **131** (not shown) may be disposed over a planarized surface **132*b*** (as shown, e.g., in FIG. **6**B) of the second encapsulant **130**, **130*b***. The additional seed layer **131** may be removed, with the exceptions of those portions of the seed layer **131** that are covered by the land pad **142** and SMS **154**. In some embodiments, the seed layer **131** may be used as the plating bus for the additional conductive layer **138** and LGA pads **142** and the SMS **154**. In additional embodiments, the method as disclosed uses one or more of the plating busses **68*a*** and the tie bars **68*e*** of conductive layer **68** as the plating bus for the conductive layers **138**, **140**, **142**, **154** and subsequent layers. Accordingly, in some instances the seed layer **131** for the LGA pads **142** and the SMS **154** is unnecessary. FIGS. **4**D_1 and **4**D_2 further illustrate where the LGA pads **142** are not encapsulant-defined and form a side surface **142*a*** on vertical sides of the LGA pads **142** and the SMS **154**.

[0084] In FIG. **4**D_2, the individual QFN, DEN or SON packages may be singulated from each other by a sawing process using saw **160**, or other suitable process including laser or scoring. As shown in FIG. **4**D_2, the singulation edge or package edge **72**, comprising cut encapsulant **156** (as shown, e.g., in FIGS. **2**, **4**A, and **8**), may be offset by first distance **68*h*** from both the conductive LGA pad **142** and the SMS **154** for a better cut. FIG. **4**D_2 further shows singulation through tie bars **68*e*** as part of traces **68*c*** to form exposed, cut tie bar ends **68*f*,** as also depicted in FIG. **4**A, among others.

[0085] FIG. **4**E_1 shows an enlargement of a further embodiment of a cross-sectional side view of a portion of FIG. **4**C indicated by detail mark **4**D. The seed layer **131**, conductive layer **68** and second encapsulant **130**, **130*b*** may be formed as shown and described for the embodiment of FIG. **4**D_1. FIG. **4**E_1 depicts a vertical conductive element **140** having second encapsulant **130**, **130*b*** disposed around and contacting vertical conductive element **140** with a pad finish or SMS **154** disposed over the vertical conductive element **140** after conductive element **140** has been exposed from second encapsulant **130**, **130*b*** by a planarization or grinding process to form planarized surface **132*b*** (as shown in FIGS. **6**A and **6**B), without the formation of a separate LGA pad **140** of FIG. **4**D_1.

[0086] FIG. **4**E_2 illustrates the cross-sectional side view of FIG. **4**E_1 during a sawing or singulation process to form individual QFN, DEN or SON packages using saw **160**, or other suitable process including laser or scoring, as similarly shown and described for FIG. **4**D_2. The singulation edge or package edge **72**, comprising cut encapsulant **156** (as shown, e.g., in FIGS. **2**, **4**A, and **8**), may be offset by first distance **68*h*** from both the conductive LGA pad **142** and the SMS **154** for a better cut. FIG. **4**E_2 further shows singulation through tie bars **68*e*** as part of traces **68*c*** to form exposed, cut tie bar ends **68*f*,** as also depicted in FIG. **4**A, among others.

[0087] FIG. **4**F, continuing from FIG. **4**E, is a cross-sectional side view of an enlarged portion of a corner of a conductive pad or LGA pad **142**, formed according to an embodiment of the present disclosure, comprising the pad surface finish or SMS **154**. The pad surface finish or SMS **154** may comprise one or more layers of matte tin, Ni—Pd—Au, Ni—Au, Au, Ag, Ni—Pd, OSP, and solder. As shown by FIG. **4**F, additional conductive layers, such as the vertical conductive element **140** may be electroplated within openings formed through the second encapsulant **130**, **130*b***.

[0088] FIG. **4**F further illustrates a particular embodiment, in which at least one layer of encapsulant **130** comprises a locking feature **133** that is locked between, or interlocked with, two layers of a land pad **142**, and for example between the RDL or conductive layer **68**, **68*c*** and the conductive pad **142**. It should be clear that multiple layers of conductive material, such as internal land pads, vias or traces as well as multiple RDL layers, can be repeated to form the vertical conductive element **140** or the land pad **142** so that multiple layers of encapsulant **130** are interlocked between at least three or even more layers of conductive material. Seed layer **131** is depicted as remaining under conductive layer **68** after an etching process to remove the seed layer.

[0089] In other embodiments, more than just two layers of the same kind of encapsulant **130** may be formed, interlocked with layers of the land pad **142** so that the conductive layers, beginning with the conductive layer or traces **68**, **68*c*,** and ending with the land pad **142**, regardless of how many layers is used, includes interlocking structures **133** extending into the layers of the same kind of encapsulant **130**. In some particular embodiments, instead of EMC, thick mold compound (TMC), or other known mold compound may be used. By interlocking the land pad **142** with encapsulant layers **130**, particularly of encapsulant layers **130** of the same kind of encapsulant materials, an interlocking structure is formed and the land pads **142** have better adhesion to the encapsulant **130**. Furthermore, the disclosed LGA structure as depicted in FIG. **4**F resolves the prior art issues of burring at package edges because the land pads **142** do not extend to the edge **72** of the package. In addition, the disclosed LGA structure reduces exposed copper at the package edge, thereby minimizing oxidation and promoting solder wettability.

[0090] FIG. **5** represents a flowchart for a method **400** of making a quad flat no-lead (QFN), dual flat no-lead (DFN) or small outline no-lead (SON) package as disclosed herein. The method **400** of FIG. **5** is supported by a number of the FIGs. included herein. Element **402** discloses forming an embedded chip panel **30** comprising a plurality of semiconductor chips **14** embedded in an encapsulant **130** and separated by saw streets **72**. Element **404** discloses forming a conductive layer **68** over the embedded chip panel **30**. Element **406** discloses forming traces **68*c*** in the conductive layer **68**, the traces **68*c*** formed over the plurality of semiconductor chips **14**, the traces **68*c*** extending beyond an edge **24** of the semiconductor chips or component **14**, and coupled with conductive studs **125** coupled to the plurality of semiconductor chips **14**. Element **408** discloses forming bussing lines **68*a*** as part of the conductive layer **68**, over saw streets **72** and between the semiconductor chips **14**. Element **410** discloses forming tie bars **68*e*** as part of the conductive layer **68**, coupled with the traces **68*c*,** bussing lines **68*a*** and plating contacts **76**. Element **412** discloses electroplating an additional conductive layer **138** over the conductive layer **68** by providing a current through the bussing lines **68*a*** and tie bars **68*e*.** Land pads **142** and interior pad **68*d*** and in some embodiments interior pad or thermal pad **144** may be formed as part of element **412**. Element **414** represents disposing a second encapsulant **130**, **130*b*** over the conductive layer **68** and over and around additional conductive layer **138**. At element **416**, the second encapsulant **130**, **130*b*** is ground or planarized to form a planar surface **132*b*** to expose portions of the conductive layer **68**, such as land pads **142**, interior pad **68***d* (and in some embodiments interior pad or thermal pad **144**) as well as plating contacts **76**. The SMS **154** may be formed according to element **418**. Element **420** represents singulation of the QFN, DFN, or SON packages.

[0091] FIGS. **6**A and **6**B depict cross section views of the conductive layer **68**, additional conductive layer **138**, and plating contact **76**, depicting the formation of LGA pads **142** over contact pads **68***b*. SMS **154** is shown as deposited over the additional conductive layer **138** that is part of planar surface **132***b*, as shown in FIG. **6**B.

[0092] FIG. **7** illustrates a QFN, DFN and SON package comprising two semiconductor chips or components **14** according to the embodiments as disclosed herein. Combining two or more semiconductor chips or components **14** provides combined functionality within a single package, according to the semiconductor chip **14** functionality. Conductive layer **68** or additional conductive layer **138** may be patterned to provide electrical coupling between multiple chips or components **14** in this configuration. In some instances, the package **100**, **20**, **300** of FIG. **7** may comprise a backside laminate **50**, similar to, or as shown in FIG. **2**.

[0093] FIG. **8** illustrates QFN, DEN and SON packages comprising more than one pad (e.g. a power pad and a ground pad), independent interior pads, thermal pads or ground planes **144**, which may be formed over additional conductive layer **138**, comprising thermal pad, or ground pad **138***a*. In some embodiments as shown in FIG. **8**, one or more land pads **142** (as shown in FIG. **4**A) may be electrically connected to interior pads, thermal pads or ground planes **144**. In some embodiments, the QFN, DEN and SON packages **100**, **200**, **300** may comprise one or more of multiple independent interior pads, land pads formed in single row, multiple-rows, land grid array (LGA), ball grid array (BGA), or as an area array for thermal, power, and signal interconnection.

[0094] This disclosure, its aspects and implementations, are not limited to the specific package types, material types, or other system component examples, or methods disclosed herein. Many additional components, manufacturing and assembly procedures known in the art consistent with semiconductor wafer fabrication, manufacture and packaging are contemplated for use with particular implementations from this disclosure. Accordingly, for example, although particular implementations are disclosed, such implementations and implementing components may comprise any components, models, types, materials, versions, quantities, or the like as is known in the art for such systems and implementing components, consistent with the intended operation.

1. A method of making a quad flat no-lead (QFN), dual flat no-lead (DFN) or small outline no-lead (SON) package, the method comprising:

forming an embedded chip panel comprising a plurality of semiconductor chips embedded in encapsulant and separated by saw streets;

forming a conductive layer over the embedded chip panel, the conductive layer comprising:

contact pads and traces formed over the plurality of semiconductor chips, the traces extending beyond an edge of the semiconductor chips and coupled with vertical conductive elements,

bussing lines formed over the saw streets, and

tie bars extending between and coupled with the contact pads and the bussing lines;

forming vertical conductive elements disposed over and coupled with the conductive layer;

disposing an encapsulant layer over the conductive layer and around the vertical conductive elements such that the vertical conductive elements comprise exposed ends, and wherein at least a portion of the tie bars comprise cut tie bar ends exposed from the encapsulant layer;

electroplating land pads, a conductive pad finish, or a solderable metal system (SMS) over the vertical conductive element by providing a current through the bussing lines and tie bars, wherein the land pads, conductive pad finish, or SMS are electroplated and protrude from the encapsulant layer; and

singulating the embedded chip panel through the saw streets to form a plurality of QFN, DFN, or SON packages with exposed ends of the tie bars exposed at a periphery of the QFN, DFN, or SON packages.

2. The method of claim **1**, wherein forming the embedded chip panel further comprises:

disposing a plurality of semiconductor chips face-up on a temporary carrier, the plurality of semiconductor chips comprising the conductive studs disposed over active layers of the plurality of semiconductor chips;

disposing encapsulant in a single step around four side surfaces of the plurality of semiconductor chips, over the active layers of the plurality of semiconductor chips, and around at least a portion of sidewalls of the conductive studs, wherein a portion of the encapsulant disposed between the plurality of semiconductor chips forms the saw street; and

planarizing a surface of the encapsulant and conductive studs.

3. The method of claim **1**, wherein interior pads are formed over the plurality of semiconductor chips.

4. The method of claim **1**, wherein the cut tie bar ends comprise a cross-sectional area in a range of 5-20 micrometers-squared.

5. The method of claim **1**, wherein electroplating the land pads, conductive pad finish, or SMS forms a side surface.

6. The method of claim **1**, wherein the QFN, DFN, or SON package comprises a total height less than 300 micrometers.

7. The method of claim **1**, wherein the QFN, DEN, or SON package comprises one or more of:

multiple independent interior pads; or

land pads formed in single row, multiple rows, or as an area array.

8. The method of claim **1**, further comprising forming the conductive layer comprising reinforcement strips formed around a periphery of the embedded chip panel.

9. The method of claim **1**, further comprising forming the QFN, DFN, or SON packages as System-in-Packages (SiPs) or Multi-Chip-Modules (MCMs) comprising two or more chips in at least one of the QFN, DFN, or SON packages.

10. The method of claim **1**, further comprising forming at least a portion of the conductive layer using unit-specific patterning to account for actual positions of the plurality of semiconductor chips embedded within the embedded chip panel.

11. A method of making a quad flat no-lead (QFN), dual flat no-lead (DFN) or small outline no-lead (SON) package, the method comprising:

forming an embedded chip panel comprising a plurality of semiconductor chips embedded in encapsulant and separated by saw streets;

forming a conductive layer over the embedded chip panel, the conductive layer comprising bussing lines, contact pads, traces, and tie bars, wherein the tie bars extend between and are coupled with the contact pads and the bussing lines, and wherein the bussing lines and tie bars extend into the saw streets;

forming vertical conductive elements disposed over and coupled with the conductive layer;

disposing an encapsulant layer over the conductive layer and around the vertical conductive elements such that the vertical conductive elements comprise exposed ends, and wherein at least a portion of the tie bars is exposed from the encapsulant layer; and

electroplating land pads, a conductive pad finish, or a solderable metal system (SMS) over the vertical conductive elements by providing a current through the bussing lines and tie bars.

12. The method of claim **11**, further comprising electroplating a pad surface finish over the land pads to form a side surface for the land pads.

13. The method of claim **11**, wherein the QFN, DEN, or SON package comprises one or more of multiple independent interior pads, land pads formed in single row, multiple-rows, land grid array (LGA), ball grid array (BGA), or as an area array for thermal, power, signal.

14. The method of claim **11**, further comprising singulating the embedded chip panel through the saw streets, comprising:

removing a portion of the embedded chip panel to sever the tie bars; and

singulating the strips into individual QFN, DFN, or SON packages.

15. The method of claim **11**, further comprising forming the QFN, DFN, or SON packages as System-in-Packages (SiPs) or Multi-Chip-Modules (MCMs) comprising two or more chips in at least one of the QFN, DFN, or SON packages.

16. The method of claim **11**, further comprising forming at least a portion of the conductive layer using unit-specific patterning to account for actual positions of the plurality of semiconductor chips embedded within the embedded chip panel.

17. A method of making a quad flat no-lead (QFN), dual flat no-lead (DFN) or small outline no-lead (SON) package, the method comprising:

forming an embedded chip panel comprising a plurality of semiconductor chips embedded in encapsulant and separated by saw streets;

forming a conductive layer over the embedded chip panel and extending into the saw street;

electroplating a vertical conductive element over the conductive layer by providing a current through the conductive layer;

disposing an encapsulant layer over the conductive layer, wherein at least a portion of the conductive layer is exposed from the encapsulant layer; and

electroplating land pads, a conductive pad finish, or a solderable metal system (SMS) over the vertical conductive element by providing a current through conductive layer.

18. The package of claim **17**, further comprising an electroplated interior pad disposed over the embedded component.

19. The package of claim **17**, further comprising:

the embedded component comprises conductive studs coupled to an active layer of the embedded component; and

encapsulant is over the active layer of the embedded component and contacts a side surface of the conductive studs.

20. The package of claim **17**, further comprising a pad surface finish over the land pads to form a side surface for the land pads that is not encapsulant defined.

21. The method of claim **17**, wherein the QFN, DFN, or SON package comprises one or more of multiple independent interior pads, land pads formed in single row, multiple-rows, land grid array (LGA), ball grid array (BGA), or as an area array for thermal, power, signal.

22. The package of claim **17**, further comprising forming the QFN, DFN, or SON packages as System-in-Packages (SiPs) or Multi-Chip-Modules (MCMs) comprising two or more embedded components in the QFN, DFN, or SON packages.

23. The package of claim **17**, further comprising unit-specific patterning that results with a misalignment between the embedded component and the conductive layer that is less than a misalignment between the embedded component and a package edge.

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